

PitStop

Master System Documentation

Complete reference: customer app, owner panels, Super Admin, database, auth, booking, and deployment.

التوثيق الشامل — عمل، صاحب محل، مدير فرع، Super Admin، قاعدة البيانات، والحجز

[Expo 54](#)[React Native](#)[Supabase](#)[Super Admin](#)[AR / EN](#)[PitStop 2.0](#)[Premium SaaS](#)[Branch GPS](#)[Walk-in POS](#)[Driver Network](#)[Moderation](#)[Realtime Offers](#)[Campaign Engine](#)

Project: car-care-eg · PitStop EG · Version 2.0 · Updated Jul 2026: booking lifecycle engine, marketing campaigns, merchant realtime, custom confirm modal, Talabat-style bookings UX, migrations 10–13 campaigns, full system flowcharts (\$40–\$46)

Table of Contents

1. Architecture Overview
2. App Routes & Screens
3. Authentication & Roles
4. Customer Features
5. Shop Types & Owner Panels
6. Booking Flow
7. Parts & Store Orders
8. Database Schema
9. Supabase Setup & Migrations
10. Key Library Modules
11. Internationalization
12. Notifications & PDF Reports
13. Demo Accounts
14. Environment & Running the App
15. Limitations & Roadmap
16. UI Architecture & Design System
17. Master Navigation Flowcharts
18. Customer UI — Module by Module
19. Owner UI — Module by Module
20. Screen-by-Screen Reference
21. Shared UI Components
22. UI State & Context Flow
23. Super Admin — Overview & Roles
24. Super Admin — Flowcharts
25. Super Admin — Modules & UI
26. Merchant Registration Module
27. New Library Modules (Admin)
28. Car Wash Customer UX (Busy · Loyalty · Progress)
29. SaaS Premium — Free vs Premium (Wash Owner)

- 30. Branch GPS & Customer Location Flow
- 31. Web Multi-Tab Auth Isolation
- 32. Wash Pricing — Services Only (No Base Price)
- 33. Walk-in POS — Quick Counter Cashier
- 34. Driver Network & Community Feed
- 35. Wash Owner vs Branch Manager (Role Split)
- 36. Super Admin SaaS — Premium · Ledger · Moderation
- 37. Branch Profile & Cover Photos
- 38. SQL Migrations Steps 7–13 (Checklist)
- 39. Web Dialogs — userAlert & userConfirm
- 40. **40. Complete System Flowcharts (2026)**
- 41. **41. Premium Splash & Home Hero Animation**
- 42. **42. Merchant Realtime & Order Notifier**
- 43. **43. Smart Booking Lifecycle & No-Show Override**
- 44. **44. Custom Confirm Modal (Web + Native)**
- 45. **45. Marketing Offers & Campaigns Engine**
- 46. **46. Merchant History, Reports & Financial Integrity**

1. Architecture Overview

نظرة عامة على البنية

Stack

Layer	Technology
Mobile / Web UI	Expo Router ~54, React Native, expo-linear-gradient, expo-image
State	React Context (auth, i18n, theme, shop catalog)
Local storage	AsyncStorage (bookings, extras, loyalty stamps, notifications, wash branches cache)
Legacy garage	SQLite via expo-sqlite (catalog, odometer — hidden tabs)
Backend	Supabase Auth + Postgres (optional; app works offline-first)
Push	Expo Notifications → shop_push_tokens (native only)
AI	OpenAI GPT-4o-mini (Assistant tab)

Provider tree (startup)

ThemePreferenceProvider → I18nProvider → AppDialogProvider → CustomConfirmProvider → ShopCatalogProvider
→ CustomerAuthProvider → ShopAuthProvider → AppBootstrap → Root Stack

There is no separate AuthProvider. AppBootstrap blocks navigation until auth is resolved and routes users by role. CustomConfirmProvider mounts CustomConfirmModal for destructive merchant confirms (no-show override) on web and native.

Data flow (hybrid)

Customer action (book, order)
→ lib/booking/storage.ts or partsStorage.ts
→ Supabase insert (if configured) + AsyncStorage cache
→ Owner notification (local + optional Expo push)
→ Owner approves/declines → customer in-app notification

Shop type enum

maintenance · wash · parts · accessories · winch

2. App Routes & Screens

Root stack (app/_layout.tsx)

Route	File	Purpose
/welcome	welcome.tsx	Customer + owner login, merchant registration, guest, language
/(tabs)/*	(tabs)/_layout.tsx	Main tab navigator
/service/[type]	service/[type]/index.tsx	Pick area (Cairo/Giza) for service type
/service/[type]/[areald]	service/[type]/[areald].tsx	Shops in area → profile or parts shop
/shop-profile/[shopId]	shop-profile/[shopId].tsx	Public profile: gallery, hours, services, reviews, book
/book/[shopId]	book/[shopId].tsx	Booking form: date, slots, car, phone, notes
/nearby/[type]	nearby/[type].tsx	GPS discovery — wash lists per branch, Haversine sort, refreshes on focus
/parts-shop/[shopId]	parts-shop/[shopId].tsx	Parts/accessories cart + checkout
/auth-required	auth-required.tsx	Gate before booking/purchase for guests
/settings/vehicles	settings/vehicles.tsx	Customer vehicle list
/shop/wash-owner-hub	shop/wash-owner-hub.tsx	Wash hub: notifications, orders, history
/admin	admin/index.tsx	Super Admin panel (4 tabs: dashboard, pending, merchants, moderation)
/booking/[id]	booking/[id].tsx	Order details, cancel, contact, rating
/points-marketplace	points-marketplace.tsx	Loyalty points marketplace (Settings link)
/driver-network/[postId]	driver-network/[postId].tsx	Single community post
/shop/merchant-*	shop/merchant-*.tsx	Owner settings, staff, services, hours, password, terms, privacy
/reset-password	reset-password.tsx	Supabase password reset deep link

Removed legacy routes: /login, /verify (phone OTP), /car/[id], /add-car, garage catalog tabs — no longer in the app.

Customer tabs (app/(tabs)/)

Tab	Visible	Purpose
index (Home)	Customer/guest	Service cards, offers, next booking
bookings	Customer/guest	My bookings, cancel, history
favorites	Logged-in customer	Saved shops
assistant	Customer/guest	Driver Network + AI chat
settings (My account)	Logged-in customer / guest	Language, theme, vehicles, points marketplace, support, logout
shop	Shop session	Owner login or owner dashboard

Legacy hidden tabs (catalog, alerts, garage) removed from customer navigation.

Navigation rules (AppBootstrap.tsx)

- Public paths: /welcome, /reset-password, /auth-required
- **Admin session** → redirect to /admin
- **Pending owner** → stay on /welcome?pending=1 (no shop access)
- Approved shop session → redirect to /shop
- No session on protected route → /welcome
- Customer on /welcome after login → home or returnTo deep link

3. Authentication & Roles

المصادقة والأدوار

Roles (`public.user_role`)

Role	Login	shop_id	branch_id	Purpose
customer	Yes	—	—	Books services, favorites, notifications
owner	Yes	Required	—	Owns brand, all branches, creates managers
branch_manager	Yes	Required	Required	Runs one branch day-to-day
pending_owner	Yes	Required	—	New merchant awaiting admin approval; <code>is_active=false</code>
admin	Yes	—	—	Super Admin → /admin panel

Employees are NOT auth users — stored in `branch_employees` only.

Shop visibility: `shops.is_active=false` hides shop from customer catalog until admin approves.

Customer auth (`CustomerAuthContext.tsx`)

- Supabase email/password: `signIn`, `signUp`, `resetPassword`
- Guest mode: local flag, no Supabase session
- Blocks shop staff emails from customer login
- Requires email confirmed for login
- Storage: `@pitstop/customer-session`, `@pitstop/guest-session`

Shop auth (`ShopAuthContext.tsx`)

- Resolves from `public.users` (`owner` | `branch_manager` | `admin` | `pending_owner`) or legacy `shops.owner_email`
- Flags: `isOwner`, `isBranchManager`, `isAdmin`, `isPendingOwner`, `isPremium`
- `registerOwner()` — `signUp` + RPC `register_shop_owner` → **signOut** (user sees success alert, logs in later when approved)
- Login results: `ok` | `ok_admin` | `pending_approval` | `invalid_credentials` | `email_not_confirmed` | `email_login_disabled` | ...
- Storage: `@pitstop/shop-session` (staff email)
- Branch manager invite uses `staffInviteLock` to prevent session hijack
- **Web login fix:** `lib/auth/authMutationLock.ts` blocks `onAuthStateChange` side effects during `signIn/signUp` to prevent Supabase auth deadlocks (infinite “Signing in...”)

Customer auth — web deadlock fix

Both `CustomerAuthContext` and `ShopAuthContext` defer Supabase queries out of `onAuthStateChange` using `setTimeout(0)` and skip listeners while `beginAuthMutation()` is active. Welcome screen shows contextual button labels: Signing in... / Creating account... / Redirecting...

إصلاح تعليق تسجيل الدخول على الويب: لا تُنفَّذ استعلامات Supabase داخل `onAuthStateChange` أثناء `signIn`

Auth storage on web (per-tab sessions)

Web uses **tab-isolated** storage so you can test owner + customer in parallel browser tabs:

- `lib/supabase/authStorage.ts` — Supabase JWT in `sessionStorage` (not `localStorage`)
- `lib/storage/webTabAuthStorage.ts` — `@pitstop/customer-session`, `@pitstop/guest-session`, `@pitstop/shop-session` in `sessionStorage`

- `lib/supabase/webTabAuthIsolation.ts` — disables Supabase BroadcastChannel cross-tab sync; `signOut({ scope: 'local' })` on web so logout in one tab does not revoke the other tab's session
- Legacy Supabase keys in `localStorage` are cleared on first client init

على الويب: كل تاب له session مستقل — owner في تاب و customer في تاب ثاني من غير ما يتزامنوا.

Role hierarchy (updated)

Super Admin (platform)

- └ Approves / rejects `pending_owner` registrations
- └ Views platform stats (bookings, 12% fee)
- └ Sees all merchants + branches

Owner (approved)

- └ Creates branches
- └ Creates / links Branch Managers
- └ Sees all branches + reports

`pending_owner` (awaiting approval)

- └ Shop hidden from customers until admin accepts

Branch Manager (one branch)

- └ Manages daily operations
- └ Adds employees (no login)

Employees → roster only (`branch_employees`)

4. Customer Features

مميزات العميل

Home (/tabs/index.tsx)

- Service entry cards: maintenance, wash, parts, accessories
- Live promotional offers from shops
- Next upcoming booking snippet
- **Active vehicle** — dropdown picker synced with Manage vehicles
- Points marketplace link from Settings (not on Home)

Car Wash customer UX (new)

- **Busy badge:** pulsing red notice when owner sets branch status to Busy — on shop list cards and shop profile
- **Loyalty stamps:** offline-first; +1 stamp when a wash booking reaches done; reward coupon at 5 stamps
- **Today's booking progress:** 3-step timeline on My Bookings for same-day appointments (Pending → Confirmed → Done). Legacy DB status `in_progress` maps to Confirmed in UI.
- **Completed booking rating:** Talabat-style 5-star footer on My Bookings when status is done — one review per customer per shop (mutually exclusive with shop profile form)

مميزات غسيل جديدة: شارة مزدحم، بطاقة ولاء 5 أختام، شريط تقدم للحجوزات اليوم.

Discovery

- **By area:** `/service/{type}` → pick Maadi, Nasr City, El Rehab, etc.
- **Nearby GPS:** `/nearby/{type}` — device location permission → Haversine distance sort (closest first). For wash, lists each active `shop_branches` row (not just shop brand). Screen reloads on focus so owner GPS updates appear without full app restart.
- **Directions:** shop profile + list cards → Google Maps deep link (<https://maps.google.com/?q=lat,lng>) using branch coordinates when available
- **Favorites:** saved shops tab (login required)
- **Wash pricing display:** customers see per-service prices on shop profile — no separate “base price” chip for wash shops

Shop profile

- Gallery, bilingual name/address, phone, directions, share
- Services list with prices and duration
- Reviews (ratings, comments, likes, owner replies)
- Offers / coupons display
- **Wash busy badge** when `washShopStatus=busy` (synced from owner branch status)
- More info section (owner-configured)
- Book appointment CTA → requires login (not guest)

My bookings

- List all bookings for customer phone/account
- **Progress timeline** for today's active wash/service bookings (3 steps: pending → confirmed → done)
- **Rate** footer on completed bookings (`OrderListCard`); **Book again** on the right
- Cancel pending/confirmed bookings

- Clear history
- In-app notifications for approval, decline, reminders

Driver Network + AI (/tabs/assistant.tsx)

- Community posts (local AsyncStorage)
- Embedded AI assistant (OpenAI — requires API key)

Settings

- Arabic / English toggle
- Light / dark theme
- Vehicle management link
- Support, privacy, sign out

5. Shop Types & Owner Panels

أنواع المحلات ولوحات التحكم

Panel routing (app/(tabs)/shop.tsx)

wash type → WashOwnerPanel.tsx (full CMS)
parts/accessories → Store owner UI (inventory + orders)
maintenance/winch → Generic owner UI (profile + bookings + PDF)

Generic owner panel (maintenance, winch)

Section	Features
Profile	Bilingual name, address, phone, email, more info; winch toggle (maintenance)
Gallery	Add/remove photos
Pricing	Default service price (EGP)
Schedule	Open/close, slot duration; Friday closed default
Offers	Bilingual promotional offers
Bookings	Active requests (confirm/cancel/done), history, date-range reports
PDF report	View report modal + save/share PDF
Notifications	Booking + parts order alerts

Store owner panel (parts, accessories)

Section	Features
Profile + gallery	Same as generic (no schedule/bookings)
Inventory	Add items: name, price, stock, image; ±1 stock
Orders	Confirm / cancel / mark shipped; 12% platform fee on orders

Wash owner panel (WashOwnerPanel1.tsx) — PitStop 2.0 + Premium

Section	Owner	Branch Manager	Premium required?
Dashboard stats (today + pending)	All branches	Own branch	No (Free)
Dashboard stats (revenue, rating, customers)	All branches	Own branch	Yes
Branch status (open/closed/busy/vacation)	Per branch	Own branch	No
Analytics widgets	Per branch	Own branch	Yes
Branch switcher + add branch	Yes	No (fixed branch)	Yes (Free = single branch only)
Profile + GPS pin + gallery	Per branch	Own branch	No
Services CRUD (each service has price)	Per branch	Own branch	Yes
Weekly hours + vacation	Per branch	Own branch	Yes
Coupons	Per branch	Own branch	Yes
Bookings workflow	All branches	Own branch	No
Employees roster	All branches	Own branch	Yes

Create branch manager	Yes	No	Yes
Reviews moderation	Yes	Yes (own branch)	Yes
PDF reports	All	Own branch	Yes

Premium gate: PremiumFeatureGate.tsx dims locked sections; tap opens PremiumUpgradeModal (EN/AR). Flag from shops.is_premium via ShopAuthContext.isPremium.

الخطة المجانية: إحصائيات أساسية + حالة الفرع + الملف + المعرض + الحجوزات. Premium: فروع متعددة، خدمات، تحليلات، كويونات، مديرين، تقارير.

Service categories (wash)

exterior_wash · interior_cleaning · full_package · detailing · engine_cleaning · custom

Wash owner hub (/shop/wash-owner-hub)

Tab	Purpose
notifications	Wash notification inbox
orders	Approve/reject/update booking statuses
history	Past bookings; clear history

6. Booking Flow

مسار الحجز من البداية للنهاية

End-to-end workflow

```
Home / Nearby / Favorites
→ Shop profile (/shop-profile/[shopId]) – promo banner if active campaign
→ [Guest?] → /auth-required → /welcome login
→ Book (/book/[shopId]?serviceId=...)
→ Active campaign banner + dynamic price (% , flat, buy-X-get-Y)
→ Select date + time slot + car + phone + notes
→ createBooking(offerId?) → status: pending
→ Owner notification (+ optional push + Supabase Realtime)
→ Owner: confirm → confirmed | decline → cancelled
→ Customer notification + reminders (60 min, 30 min before)
→ [Auto lifecycle] confirmed + scheduled_at > 1h ago → virtual done at read time
→ Owner: Complete → done | No-show override → no_show (CustomConfirmModal)
→ Customer: optional 5-star rating on My Bookings (one per shop)
```

Virtual auto-done lifecycle (lib/booking/storage.ts)

If a booking is confirmed or in_progress and its scheduled_at is more than **1 hour** in the past, read-time helpers treat it as done with flag lifecycleAutoCompleted: true. This prevents stale confirmed rows from blocking merchant history. Merchants can override auto-completed rows to no_show via promptMerchantNoShowOverride(). See §43.

Booking statuses (UI)

pending → confirmed → done | cancelled | no_show

There is no separate “In progress” step in the customer UI or owner wash buttons. Legacy rows with in_progress in the database still display as **Confirmed** and map to step 2 of the 3-step timeline.

Pricing (lib/booking/storage.ts)

Shop type	Booking price source
wash	Selected service priceEgp from branch menu; fallback = first active service (no owner “base price” UI)
maintenance	Owner default service price in shop extras, or type default 650 EGP
winch	Type default 500 EGP
other	Type default 420 EGP

Platform fee: **12%** of service price (shown in owner PDF reports).

Time slots

Built by buildSlotsForShopDate() from owner weekly hours + existing bookings. Submit disabled until a slot is selected.

Persistence

- Primary: Supabase bookings table (when configured)
- Cache: AsyncStorage @pitstop/bookings/v1
- On Supabase insert failure → error shown to customer (RLS fix: step4 SQL)

7. Parts & Store Orders

طلبات قطع الغيار والإكسسوارات

```
/parts-shop/[shopId]
  → Browse inventory (store table)
  → Cart + shipping address + phone
  → createPartsOrder()
  → Owner notification + push
  → Owner: confirm / cancel / ship
  → Customer notification
```

Order statuses: pending → confirmed → shipped | cancelled

8. Database Schema

هيكل قاعدة البيانات

Entity diagram

```
auth.users (Supabase Auth)
└─ public.users (role, shop_id, branch_id)

public.shops (brand)
├─ shop_branches (locations, hours, gallery, status)
│   └─ branch_services
│       └─ branch_employees (no login)
├─ bookings (+ branch_id, service_id)
├─ store (parts/accessories inventory)
└─ parts_orders + parts_order_items

shop_push_tokens · garage_snapshots (legacy) · areas
```

Core tables

Table	Purpose
areas	Cairo/Giza districts + e1-rehab (El Rehab City demo)
shops	Brand catalog; is_premium; platform_fee_last_settled_at for admin ledger
users	App identity: customer, owner, branch_manager, admin
shop_branches	Physical branches: profile, weekly_hours, vacation_mode, branch GPS latitude/longitude
branch_employees	Staff roster (full_name, phone, job_title, notes)
branch_services	Per-branch service menu
bookings	Appointments; booking_type app walk_in; status, pricing, branch_id, offer_id
offers	Marketing campaigns: offer_type percentage flat_amount buy_x_get_y; discount_value; required_wash_count; expires_at; Realtime enabled
shop_reviews	Customer reviews; owner_reply; hidden; reported flags
posts · comments · post_votes · comment_likes	Driver Network community feed
store	Parts/accessories SKUs with stock_qty
parts_orders	Customer store orders
shop_push_tokens	Expo push tokens per shop

RLS helpers

is_shop_owner · is_branch_manager · can_manage_shop · can_manage_branch

9. Supabase Setup & Migrations

إعداد Supabase وملفات ال SQL

Fresh project

1. supabase/schema.sql
2. supabase/seed.sql — 12 demo shops
3. supabase/seed-branches.sql — optional 2nd branch + employees

Existing project (PitStop 2.0 migration)

1. apply-pitstop-2.0-step1-enums.sql — run alone first
2. apply-pitstop-2.0-step2-main.sql — users, branches, employees
3. apply-pitstop-2.0-step3-branch-manager-rpc.sql — assign_branch_manager RPC
4. apply-pitstop-2.0-step4-bookings-rls.sql — restore booking insert policies
5. apply-pitstop-2.0-step5-admin-approval.sql — Super Admin, pending_owner, approve/reject RPCs
6. apply-pitstop-2.0-step6-shop-premium.sql — shops.is_premium boolean default false
7. apply-pitstop-2.0-steps-7-through-8-bundle.sql — walk-in bookings, shop_reviews, community feed (posts, comments, likes)
8. apply-pitstop-2.0-step7b-walk-in-pos-refactor.sql — optional alone: resolve_customer_id_by_phone() RPC
9. apply-pitstop-2.0-step9-admin-saas.sql — admin premium RPC, merchant ledger settlement, moderation flags
10. seed-admin-user.sql — link Auth user as platform admin in public.users
11. seed-el-rehab-wash.sql — El Rehab area + shop-wash-rehab branch at 30.0244, 31.4939 + owner link (requires Auth user rehab.wash@demo.com first)
12. SETUP-AUTH-CHECKLIST.md — step-by-step auth setup (Email provider, demo accounts)
13. seed-test-data.sql — optional demo manager link

Other migration files

- migrate-roles-branches.sql — legacy → 2.0
- migrate-to-store.sql — spare_parts → store
- apply-pitstop-2.0-on-existing-db.sql — combined path

Step 5 RPCs (Super Admin)

RPC	Caller	Effect
register_shop_owner(...)	New merchant (authenticated)	Creates inactive shop + main branch; user → pending_owner
approve_shop_owner(user_id, shop_id)	Admin	user → owner, is_active=true; shop → is_active=true (live)
reject_shop_owner(user_id, shop_id)	Admin	Deletes shop; user → customer
admin_toggle_shop_premium(shop_id, is_premium)	Admin	Instantly toggles shops.is_premium (after confirm dialog)
admin_settle_shop_platform_fees(shop_id)	Admin	Sets platform_fee_last_settled_at = now() — resets ledger debt

resolve_customer_id_by_phone(phone)	Walk-in POS	Links counter phone to app customer UUID for loyalty
-------------------------------------	-------------	--

Link owner after Auth user created

```
update public.users
set role = 'owner', shop_id = 'shop-wash-nile'
where email = 'wash@demo.com';
```


10. Key Library Modules

أهم ملفات /lib

Module	Role
lib/booking/storage.ts	Bookings CRUD, virtual lifecycle (auto-done), Supabase + AsyncStorage merge, campaign pricing on create
lib/booking/offerRepository.ts	deployShopCampaign(), listActiveOffersForShop(), subscribeOffersRealtime(), countDoneBookingsForCustomerAtShop()
lib/booking/offerPricing.ts	applyCampaignPrice(), formatOfferBadge(), pickBestLiveOffer(), isOfferLive()
lib/booking/merchantBookingOverride.ts	promptMerchantNoShowOverride() → CustomConfirmProvider
lib/booking/bookingHistoryRepository.ts	Archived history for merchant; isFinalizedHistoryBooking filter
lib/notifications/notificationService.ts	Merchant booking Realtime channel, pending queue, mergeBookingList + lifecycle
components/merchant/OrderNotifier.tsx	useMerchantOrderNotifier hook — patchBookingLocally, pending count
components/merchant/MerchantCampaignsPanel.tsx	Deploy + list active campaigns (Form + List)
components/common/CustomConfirmModal.tsx	Themed destructive confirm dialog (replaces window.confirm)
lib/ui/CustomConfirmProvider.tsx	Global showCustomConfirm() imperative API
components/PremiumAnimatedSplash.tsx	Branded splash overlay (#080D1A / #00D4FF)
components/home/HomeHeroCarAnimation.tsx	Idle + drive SVG animation on customer Home
components/owner/OwnerHistoryPanel.tsx	History filters, PDF reports, no-show override on auto-completed rows
lib/booking/catalogRepository.ts	Areas/shops from Supabase + cache
lib/booking/shopExtrasStorage.ts	Owner profile, schedule, services, offers, washShopStatus sync
lib/booking/loyaltyStampsStorage.ts	Offline loyalty stamps + reward coupons (@pitstop/loyalty_stamps)
lib/booking/shopSchedule.ts	Weekly hours, slot builder, open-now, busy status labels
lib/booking/reporting.ts	Date ranges, EGP, buildOwnerReportHtml()
lib/booking/commerceEvents.ts	Owner/customer notifications
lib/booking/bookingReminders.ts	60/30 min reminders (native)
lib/booking/partsStorage.ts	Store inventory + orders
lib/booking/wash/washBranchStorage.ts	Multi-branch state (Supabase-first + cache)
lib/booking/nearby.ts	Haversine distance, formatDistanceAway, shop sort helpers
lib/booking/washBranchNearby.ts	Lists wash branches from Supabase, sorts closest-first per branch pin
lib/linking/contact.ts	Phone, maps, openBranchDirections(), openMapsAtCoordinates()

lib/booking/wash/branchRepository.ts	Branch CRUD, GPS update, resolveRemoteBranchId, fetchDefaultBranchCoordinates
lib/booking/wash/branchManagerRepository.ts	Manager account RPC + signUp flow
lib/shop/ownerRegistrationRepository.ts	Merchant signUp + register_shop_owner RPC
lib/admin/adminRepository.ts	Pending, merchants, stats, ledger, moderation, premium toggle, approve/reject
components/admin/AdminPanel.tsx	Super Admin UI (4 tabs + confirm dialogs)
lib/community/postRepository.ts	Driver Network posts, comments, likes, report flags
components/owner/wash/WalkInBookingModal.tsx	Quick Counter Cashier (POS) walk-in form
lib/booking/reviewRepository.ts	Shop reviews sync + owner reply notifications
lib/shop/shopStaffUser.ts	Resolve owner/manager/admin/pending from users
lib/supabase/client.ts	Supabase singleton; calls prepareWebTabAuthIsolation() on init
lib/supabase/authStorage.ts	sessionStorage (web) / AsyncStorage (native) for Supabase JWT
lib/supabase/webTabAuthIsolation.ts	Disable BroadcastChannel; local signOut; clear legacy localStorage keys
lib/storage/webTabAuthStorage.ts	Per-tab customer/guest/shop session keys on web
components/owner/PremiumFeatureGate.tsx	Locks premium wash owner sections
components/owner/PremiumUpgradeModal.tsx	Upgrade CTA modal (EN/AR)
lib/push/shopPush.ts	Expo push for owners (native)
lib/i18n/strings.ts	~928 translation keys × EN + AR
lib/auth/authMutationLock.ts	Prevents Supabase auth deadlock during signIn/signUp
components/ui/WashBusyBadge.tsx	Pulsing red busy notice for customers
components/customer/ActiveVehiclePicker.tsx	Home + booking — active vehicle dropdown synced with Manage vehicles
components/customer/OrderListCard.tsx	My Bookings card — timeline, rate footer, book again
components/customer/BookingProgressTimeline.tsx	3-step booking progress for today
lib/ui/userAlert.ts	userAlert() + userConfirm() — web-safe dialogs
lib/ai-chat.ts	OpenAI assistant

11. Internationalization

العربية والإنجليزية

- Locales: en | ar — stored in @pitstop/locale
- Default from device language (expo-localization)
- API: `t(key)`, `tp(key, vars)`, `isRTL`
- RTL: `I18nManager` (native) + `document.dir` (web)
- Toggle: Welcome screen + Settings

12. Notifications & PDF Reports

الإشعارات وتقارير PDF

Owner push (native only)

- Table: shop_push_tokens
- Triggers: new booking, new parts order
- Via Expo Push API

Merchant booking Realtime (lib/notifications/notificationService.ts)

```
Supabase channel: public:bookings:{shopId}  
  INSERT pending booking → onPendingInsert → triggerMerchantOrderAlert + pending queue  
  UPDATE any status → onBookingUpdate → mergeBookingList + applyVirtualBookingLifecycle  
  Teardown on unmount prevents double-subscribe crash  
  Integrated in: WashOwnerPanel, wash-owner-hub, shop.tsx via useMerchantOrderNotifier
```

Publication: ALTER PUBLICATION supabase_realtime ADD TABLE public.bookings; (applied on remote project)

Offers Realtime (lib/booking/offerRepository.ts)

```
Supabase channel: public:offers:customer  
  INSERT/UPDATE/DELETE on public.offers  
  → subscribeOffersRealtime(onChange)  
  → Home carousel (app/(tabs)/index.tsx) reloads liveOffers instantly
```

Publication: apply-pitstop-2.0-step13-campaigns.sql adds offers to Realtime.

Customer in-app notifications

Kinds: booking_approved, booking_declined, booking_reminder, parts_order_confirmed, parts_order_cancelled

PDF reports

- buildOwnerReportHtml() — shop header, date range, gross/fee/net
- Web: scrollable modal preview → Save as PDF (browser print)
- Native: printToFileAsync + share
- Used in generic shop panel + WashOwnerPanel

13. Demo Accounts

حسابات تجريبية — كلمة السر: demo123

Platform admin

admin@demo.com / demo123

1. Supabase → Authentication → Providers → Email → **Enable Email provider**
2. Add user with **Auto Confirm User** checked
3. Run seed-admin-user.sql (or step5 + manual UPDATE)
4. Login: /welcome → Owner tab → redirects to /admin

See supabase/SETUP-AUTH-CHECKLIST.md for full troubleshooting.

```
update public.users set role = 'admin', is_active = true, shop_id = null
where email = 'admin@demo.com';
```

Shop owners (pre-approved in seed)

Email	Shop ID	Type
wash@demo.com	shop-wash-nile	wash (set is_premium=true in SQL to test Premium UI)
rehab.wash@demo.com	shop-wash-rehab	wash — El Rehab GPS demo (create Auth user + run seed-el-rehab-wash.sql)
maintenance@demo.com	shop-maint-autofix	maintenance
parts@demo.com	shop-parts-nasr	parts
accessories@demo.com	shop-accessories-maadi	accessories
winch@demo.com	shop-winch-maadi	winch

Also: wash2@, wash3@, maintenance2@, maintenance3@, parts2@, accessories2@, winch2@ — see seed.sql

Branch manager

manager.wash@demo.com / demo123 — Nile Auto Wash main branch

Customers

customer@demo.com · sara@demo.com / demo123

14. Environment & Running the App

تشغيل المشروع

Environment variables (.env)

```
EXPO_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
EXPO_PUBLIC_SUPABASE_ANON_KEY=your-anon-key
EXPO_PUBLIC_OPENAI_API_KEY=sk-... # for AI assistant
```

Fallback (app.json extra)

Supabase URL and anon key also in app.json → extra if .env unset.

Commands

```
cd car-care-eg
npm install
npm start          # Expo (offline mode)
npm run web        # http://localhost:8082
npm run android
npm run ios
```

Supabase project (demo)

URL: <https://qlopvpyeawauepsyrirz.supabase.co>

15. Limitations & Roadmap

القيود والتطوير المستقبلي

Current limitations

Limitation	Detail
Hybrid storage	Many features use AsyncStorage; Supabase when configured
Guest cannot book	Login required despite guest browsing
Merchant approval	New owners need admin approve before shop goes live
Loyalty coupons	Generated locally; not yet validated at checkout in book flow
Winch not on home	In DB but no home service card
Web push disabled	Owner push + reminders native only
Payments	No online payment integration yet
Legacy garage	Parallel SQLite app (hidden tabs)

Future enhancements (from docs)

- Online payments (Paymob/Fawry)
- Apply loyalty coupon codes at booking checkout
- Subscriptions
- Driver network on Supabase + moderation
- Vehicle cloud sync
- Multi-region (GCC expansion)
- Assign employees to bookings/shifts
- QR check-in, live queue ETA, fleet management

Implemented: loyalty stamps, today booking progress, wash busy badge, Super Admin, Premium SaaS tiers, branch GPS sync, web multi-tab auth, wash service-only pricing.

Related documents in repo

- docs/SCHEMA.md — database reference
- docs/FUTURE_ENHANCEMENTS.md — product roadmap
- docs/pitstop-2.0-owner-srs.html — Owner/Branch SRS
- pitstop2.0.pdf — Customer-side SRS (original)

16. UI Architecture & Design System

بنية الواجهة ونظام التصميم

Layout model

```
app/_layout.tsx (Root Stack)
├─ welcome, auth-required, book, shop-profile, nearby, parts-shop ...
└─ (tabs)/_layout.tsx (Bottom tabs – customer OR shop mode)
    ├─ index (Home)
    ├─ bookings | favorites | assistant | settings ← customer area
    └─ shop ← owner dashboard (tabs hidden when shop logged in)
```

Two UI modes

Mode	Trigger	Tabs shown	Default route
Customer / Guest	No shop session	Home, Bookings, Favorites*, Assistant, Settings	/ (Home)
Shop staff	ShopAuthContext.shop set	Shop tab only (bar hidden)	/shop → AppBootstrap forces /shop

*Favorites tab hidden for guests — login required.

Theme (constants/Theme.ts + ThemePreferenceContext)

- **Light / Dark** — user choice in Settings; persisted
- **Tokens:** bg, bgElevated, card, border, text, textMuted, accent, warm, onAccent
- **RTL:** Arabic flips layout via I18nContext + I18nManager
- **Owner wash accent:** warm color in dark mode for owner buttons

UI patterns used everywhere

Pattern	Where	Behavior
Pressable + gradient btn	welcome.tsx	Login/register; pointerEvents none on gradient (web fix)
OwnerSectionCard	WashOwnerPanel	Collapsible section with title + subtitle
Modal confirm	shop.tsx, WashOwnerPanel	Replaces Alert.alert on web for booking decisions
userAlert	welcome, book, shop, admin	One-button messages: window.alert on web
userConfirm	AdminPanel	Two-button confirm: window.confirm on web (Premium, Settle)
userConfirm	AdminPanel	Cancel/OK: window.confirm on web (Premium toggle, Settle Balance)
ScrollView sections	All owner panels	Long vertical CMS layout
iframe report preview	shop.tsx, WashOwnerPanel	Scrollable HTML report before PDF

17. Master Navigation Flowcharts

مخططات التنقل الرئيسية

17.1 App entry & auth routing

```
[App launch]
|
v
[AppBootstrap waits: CustomerAuth + ShopAuth ready]
|
+-- isAdmin? -----> router.replace('/admin')
|
+-- isPendingOwner? -----> /welcome?pending=1 (blocked from /shop)
|
+-- approved shop session? --> router.replace('/shop')
|
+-- no session + protected? --> /welcome
|
+-- customer/guest on /welcome? --> / or returnTo
|
v
[/ (tabs) or stack screens]
```

17.2 Customer booking journey

```
Home (/ (tabs)/index)
| tap service card (maintenance|wash|parts|accessories)
v
/service/[type] → pick area (Maadi, Nasr City...)
v
/service/[type]/[areaId] → shop list
|
+-- parts/accessories --> /parts-shop/[shopId]
|
+-- maintenance/wash --> /shop-profile/[shopId]
|
| [Book button]
v
guest? --> /auth-required --> /welcome?returnTo=/book/...
|
v
/book/[shopId]?serviceId=...
| fill phone, car, date, time slot, notes
v
createBooking() --> success modal --> /bookings tab
```

17.3 Alternative discovery paths

```
Home --search/filter--> service cards
Home --live offers--> shop-profile
Nearby GPS (/nearby/[type]) --> shop-profile | parts-shop
Favorites tab --> shop-profile (login required)
Shop profile: Call | Directions | Share | Favorite | Gallery | Reviews
```

17.4 Owner login & panel routing

```
/welcome → Owner tab
|
```

```

+-- Login (approved owner/manager)
|   v
|   ShopAuthContext.login() → ok
|   v
|   router.replace('/shop') → WashOwnerPanel | store UI | generic UI
|
+-- Login (admin)
|   v
|   router.replace('/admin') → AdminPanel
|
+-- Login (pending_owner)
|   v
|   pending_approval message – no shop access
|
+-- Register as merchant
|   v
|   registerOwner() → signUp + register_shop_owner RPC
|   v
|   pending_owner + shop is_active=false → wait for admin

```

17.5 Wash owner hub (secondary navigation)

```

WashOwnerPanel → Hub shortcuts
  v
/shop/wash-owner-hub?tab=notifications|orders|history
|
+-- notifications: wash notification inbox
+-- orders: approve/reject/complete bookings (Accept → Complete, no In Progress button)
+-- history: completed/cancelled list + clear

```

17.6 Branch manager vs owner (wash UI)

```

Same login form (/welcome Owner tab)
  v
ShopAuthContext resolves role
|
+-- isOwner --> branch tabs visible, create branch, create manager
|
+-- isBranchManager --> single branch, no manager section, bookings filtered by branch_id

```

18. Customer UI — Module by Module

واجهة العميل — شرح كل جزء

Module A — Welcome / Auth (app/welcome.tsx)

UI elements: Logo, language toggle, Customer | Owner mode tabs, guest button, form box.

Customer tab: Login ↔ Register toggle, email/password, optional vehicles on register, forgot password link.

Owner tab: Shop email + password, demo accounts hint, merchant registration link.

Actions: onCustomerSubmit → login/register; onOwnerSubmit → shop login (owner or admin); continueAsGuest → home without Supabase.

Feedback: formMessage inline (red) + userAlert on errors; button shows Signing in... / Creating account... / Redirecting... (not stuck on "Working...").

Admin routing: ok_admin → /admin; pending owner → blocked message on welcome.

شاشة الدخول: تسجيل عميل/صاحب محل، ضيف، وإعادة توجيه الأدمن — مع إصلاح تعليق تسجيل الدخول على الويب.

Module B — Home (app/(tabs)/index.tsx)

Sections top→bottom:

1. Header logo + sign out (if logged in)
2. **Active vehicle picker** (ActiveVehiclePicker) — dropdown synced with Settings → Manage vehicles
3. Car profile card — saved make/model from active vehicle
4. Next booking card — upcoming appointment or empty state
5. Live offers carousel — wash/maintenance promotions
6. Service search + filter (all / wash only)
7. 4 service cards → navigate to /service/{type}

Loyalty points: Settings → /points-marketplace (not on Home). Wash stamp rewards still sync when owner marks booking done.

الرئيسية: اختيار السيارة النشطة + حجز القادم + العروض. نقاط الولاء من الإعدادات.

Module C — Area & Shop List (app/service/)

Step 1: Grid of Cairo/Giza areas with shop count.

Step 2: Shops in selected area; card shows name, rating, distance hint, open status.

Wash busy badge: on ShopListCard, when shop type is wash and extras washShopStatus === 'busy', shows pulsing red WashBusyBadge (compact pill).

Routing: parts/accessories → parts-shop; others → shop-profile.

قائمة المحلات: شارة «مزدحم» حمراء نابضة لمحلات الغسيل عندما يضبط المالك الحالة Busy.

Module D — Shop Profile (app/shop-profile/[shopId].tsx)

UI block	User action
Hero image + gallery swipe	View photos
Name, address, phone	Call, open maps

Schedule line	See open hours
Busy notice	WashBusyBadge full-width when wash + busy status
Services list	Owner-configured only — empty until owner adds; tap service → book with serviceId
Offers	Read promotions
More info	Owner free-text details
Reviews	Real reviews only; submit if not already rated from My Bookings
Share / Favorite	Social share; save shop
Book CTA (sticky)	→ /book or auth-required

Module E — Booking Form (app/book/[shopId].tsx)

Field	Source	Validation
Phone	Customer profile	Egypt E.164
Car type / color	Vehicle storage or manual	Car type required
Date picker	BookingDatePicker	Owner schedule days
Time slots grid	buildSlotsForShopDate	Required; booked slots disabled
Service summary	Selected service or shop default	Price + duration shown
Notes	Free text	Optional
Submit	createBooking	Disabled if no slot

Success: Modal → View my bookings → /bookings tab.

Module F — My Bookings (app/(tabs)/bookings.tsx)

OrderListCard per booking: shop name, service, datetime, price, View details link, cancel (pending/confirmed).

Today progress timeline: 3 steps (pending → confirmed → done) via BookingProgressTimeline for same-day active bookings.

Completed bookings: Talabat-style **Rate** row (5 cyan stars) when status === 'done'; one review per customer per shop. Right action is always **Book again**.

Clear history action for done/cancelled items. Detail screen: /booking/[id].

حجوزاتي: 3 خطوات لليوم + تقييم نجوم للحجوزات المكتملة + احجز تاني.

Module G — Favorites, Assistant, Settings

Favorites: Saved shop cards → shop-profile. Login required.

Assistant: Driver Network posts (local) + AI chat panel (OpenAI key required).

Settings: Language EN/AR, theme light/dark, vehicles link, support links, privacy, sign out.

Vehicles: /settings/vehicles — add/edit/delete customer cars.

Module H — Parts Shop (app/parts-shop/[shopId].tsx)

Product grid, cart badge, qty +/-, shipping address, phone, checkout → createPartsOrder.

No time slots — inventory-based e-commerce flow.

Module I — Nearby (app/nearby/[type].tsx)

Requests GPS permission → sorts shops by distance.

Filter chips: All, Top rated, Price, Distance, Open now, Favorites.

Module J — Customer Notifications (CustomerNotificationsBell)

Bell icon in tab headers (logged-in customer only).

Shows booking approved/declined/reminder + parts order updates.

Processes due reminders on focus.

19. Owner UI — Module by Module

واجهة صاحب المحل — شرح كل جزء

19.1 Generic owner panel (app/(tabs)/shop.tsx — maintenance / winch)

Header bar

Profile cover image, notifications bell, logout. Shop name from catalog.

Profile section

Fields: profile name EN/AR, address EN/AR, phone, email, more info EN/AR.

Maintenance only: Winch service toggle (shows winch add-on to customers).

Save profile info button → shopExtrasStorage.

Photo album

Add image URL, remove image. Gallery shown on shop-profile.

Service price + schedule

Default price EGP, open/close time, slot duration (15+ min). Friday closed default.

Generates customer time slots on book screen.

Offers

Add bilingual offer title/body, expiry. Cancel active offers.

Bookings dashboard

Report: date from/to, quick last N days, preview list, View report modal, Save PDF.

Active requests: pending + confirmed — Confirm | Cancel | Mark done buttons via confirm modal.

History: done + cancelled — show/hide toggle.

19.2 Store owner panel (parts / accessories)

Same profile + gallery as generic. **No schedule/bookings.**

Inventory table: add item (name, price, stock, image URL), +/- stock buttons.

Orders table: customer phone, items, total, status — Confirm | Cancel | Shipped.

19.3 Wash owner panel (WashOwnerPanel.tsx) — section order

#	Section (i18n key)	UI controls	Owner only?
1	wash_dashboard_*	Stat cards: today bookings, pending, revenue, rating, customers, top service	Stats scope differs
2	wash_status_*	Open / Closed / Busy / Vacation + vacation dates	No
3	wash_analytics_*	Revenue & booking charts (computed locally)	No

4	wash_hub_*	3 shortcut buttons → wash-owner-hub tabs	No
5	wash_branch_*	Branch tabs, add branch, rename	Owner only
6	wash_profile_*	Branch profile fields + GPS capture button (syncs to Supabase immediately) + save	No
7	wash_gallery_*	Image URL list add/remove	No
8	wash_services_*	Service list, add/edit modal (EN/AR, price, duration, category, visible, sort)	Premium
9	wash_hours_*	Day picker 0–6, open/close/break, copy to all days	Premium
10	wash_coupons_*	Coupon list, add percent/fixed code	Premium
11	wash_manager_*	Link email Create login (name, email, password)	Owner + Premium
12	wash_employees_*	Employee list per branch, add/edit/delete roster	Premium
13	wash_reviews_*	Hide review, reply, report	Premium
14	wash_report_*	Date range report + PDF (same engine as generic)	Premium
15	Active bookings inline	Status workflow: accept, reject+note, in progress, done, no show	Branch filtered · Free

19.4 Wash owner hub screens

wash-owner-hub.tsx

Tab: notifications → list from washNotificationCenter (unread badge)

Tab: orders → booking cards + action buttons + call customer

Tab: history → past bookings + delete/clear

19.5 Super Admin panel (AdminPanel.tsx)

Tab	Module	Actions
Dashboard	Platform stats cards	Read-only metrics from bookings + users
Pending	PendingOwnerRequest list	Accept / Reject → Supabase RPC
Merchants	ActiveMerchant list	View approved owners + branch count

Entry: /welcome Owner tab → admin login → /admin. Uses OwnerSectionCard for sections.

20. Screen-by-Screen Reference

مرجع سريع لكل شاشة

/welcome Auth gateway. Customer register/login, guest, owner login. Language toggle.	/(tabs)/index Home hub. Loyalty card, services, offers, next booking, car profile.
/service/[type] Pick geographic area.	/service/[type]/[areaid] Shop list for area.
/shop-profile/[shopId] Public shop page: services + prices, directions, busy badge, book.	/book/[shopId] Appointment form + slot picker.
/bookings tab Customer appointments + today progress timeline.	/favorites tab Saved shops (login).
/assistant tab Driver network + AI chat.	/settings tab App preferences.
/settings/vehicles Customer car list CRUD.	/nearby/[type] GPS shop discovery.
/parts-shop/[shopId] Storefront + cart checkout.	/auth-required Login gate with returnTo.
/(tabs)/shop Owner dashboard (type-specific panel).	/shop/wash-owner-hub Wash notifications/orders/history.
/admin Super Admin: dashboard, pending merchants, active merchants.	/reset-password Supabase password reset handler.

Tab visibility matrix

Tab	Guest	Customer	Shop owner
Home	✓	✓	hidden
Bookings	✓	✓	hidden
Favorites	hidden	✓	hidden
Assistant	✓	✓	hidden
Settings	✓	✓	hidden
Shop	hidden*	hidden	✓ (only tab)

*Shop tab visible when no customer area and not logged in as customer — rare edge case.

21. Shared UI Components

المكونات المشتركة

Component	Path	Used for
AppBootstrap	components/AppBootstrap.tsx	Auth routing, splash hide
OwnerSectionCard	components/owner/OwnerSectionCard.tsx	Wash owner CMS sections
ServiceOptionCard	components/ui/ServiceOptionCard.tsx	Home service type cards
BookingDatePicker	components/ui/BookingDatePicker.tsx	Book screen date selection
CustomerNotificationsBell	components/customer/CustomerNotificationsBell.tsx	In-app notification dropdown
WashBusyBadge	components/ui/WashBusyBadge.tsx	Pulsing red busy notice (shop list + profile)
ActiveVehiclePicker	components/customer/ActiveVehiclePicker.tsx	Home vehicle dropdown
OrderListCard	components/customer/OrderListCard.tsx	My Bookings card + rating footer
BookingProgressTimeline	components/customer/BookingProgressTimeline.tsx	3-step booking progress (My Bookings today)
ShopCatalogProvider	context/ShopCatalogContext.tsx	Loads areas/shops cache for UI lists
userAlert / userConfirm	lib/ui/userAlert.ts	Cross-platform alerts; confirm uses window.confirm on web

Context → UI binding

Context	UI reads	UI effect
I18nContext	t(), locale, isRTL	All labels; remount tabs on locale change
ThemePreferenceContext	useAppTheme()	Colors on every screen
CustomerAuthContext	customer, isGuest, busy	Auth gates, profile prefill
ShopAuthContext	shop, isOwner, isBranchManager, isPremium	Panel type, branch tabs, premium gates
ShopCatalogContext	ready, version	Shop lists, home offers

22. UI State & Context Flow

تدفق الحالة بين الواجهة والبيانات

22.1 Customer books — UI ↔ data

```
[Book screen state: phone, carType, dateYmd, timeSlot, notes]
|
| onSubmit
v
createBooking() → Supabase bookings + AsyncStorage cache
|
+→ pushOwnerNotification (owner bell)
+→ pushWashCenterNotification (wash hub)
+→ sendShopPushForBooking (native push)
|
v
[Success modal] → user taps → /bookings tab reloads listBookingsForPhone()
```

22.2 Owner confirms — UI ↔ data

```
[shop.tsx or wash-owner-hub booking card]
|
| Confirm button → confirm modal
v
updateBookingStatus(id, 'confirmed')
|
+→ pushCustomerNotification('booking_approved')
+→ scheduleBookingReminders (60m, 30m)
|
v
[CustomerNotificationsBell shows approval]
[Bookings tab status badge updates on refresh]
```

22.3 Wash branch switch — UI ↔ data

```
[Owner taps branch tab in WashOwnerPanel]
|
v
setActiveBranch(branch) + useEffect reload
|
+→ loadBranchEmployees(branch.id) from Supabase
+→ applyBranchToForms() fills profile/gallery/hours fields
|
v
[All section forms show selected branch data]
[Save writes back to washBranchStorage → branchRepository Supabase sync]
```

22.4 Report PDF — UI ↔ data

```
[Owner sets date range + Apply]
|
v
filterBookingsByDateRange() from listBookingsForShop()
|
v
buildOwnerReportHtml() → HTML string
|
```

+— Web: setReportPreviewHtml → Modal iframe (scroll) → Save PDF = print
+— Native: Print.printToFileAsync → Sharing.shareAsync

22.5 Local vs cloud (what UI shows when offline)

UI area	Offline behavior
Shop lists	Cached catalog from last Supabase refresh
Bookings list	AsyncStorage merge; may differ from cloud
Owner extras (schedule, offers)	Local shopExtrasStorage until synced
Wash branches	Local cache + Supabase when online
Loyalty stamps	Fully offline AsyncStorage; stamps on wash booking done
Wash busy status	Owner sets in panel → shopExtrasStorage → customer badge
Login	Requires Supabase (shows not_configured if missing)
Admin panel	Requires Supabase + step5 RPCs; read-only offline

22.6 Loyalty stamp — UI ↔ data

```
[Owner marks wash booking → done]
  |
  v
updateBookingStatus() in lib/booking/storage.ts
  |
  v
recordWashBookingDone(booking, previousStatus)
  |
  +→ AsyncStorage @pitstop/loyalty_stamps (per customer id/phone)
  +→ +1 stamp if wash + first time for booking id
  +→ at 5 stamps: generate LOYAL-XXXXXX coupon, reset stamps to 0
  |
  v
[Customer opens Home tab]
  |
  v
syncLoyaltyFromBookings() backfills any missed done bookings
consumePendingLoyaltyReward() → reward modal with coupon code
```

22.7 Wash busy status — owner ↔ customer

```
[WashOwnerPanel → Branch status → Busy]
  |
  v
saveWashBranchStatus() → washBranchStorage + syncWashBranchToShopExtras()
  |
  v
shopExtrasStorage.washShopStatus = 'busy' (preserved in normalizeExtras)
  |
  v
ShopCatalogContext refresh → ShopListCard + shop-profile read extras
  |
  v
[Customer sees WashBusyBadge – pulsing red notice]
```

23. Super Admin — Overview & Roles

لوحة Super Admin — نظرة عامة

PitStop 2.0 adds a **merchant approval gate**: new shop owners cannot go live until a platform admin approves them. Existing seed owners (e.g. wash@demo.com) remain pre-approved.

State machine — merchant account

State	users.role	users.is_active	shops.is_active	Customer catalog	App route
New registration	pending_owner	false	false	Hidden	/welcome (pending message)
Approved	owner	true	true	Visible	/shop
Rejected	customer	true	deleted	—	/ (customer area)

Admin capabilities

- View platform-wide booking stats and 12% platform fee estimate
- **Merchant Ledger** — per-shop outstanding platform fees since last settlement
- **Settle Balance** — confirm cash received → reset fee watermark
- List pending merchant registration requests → Accept / Reject RPCs
- List active merchants with branch counts + **Premium Tier toggle** (with confirmation dialog)
- **Moderation tab** — reported Driver Network posts/comments + flagged shop reviews

24. Super Admin — Flowcharts

مخططات Super Admin والتسجيل

24.1 Merchant self-registration

```
/welcome → Owner tab → "Register as merchant"
|
v
Fill: business name, owner name, phone, address, shop type, area, email, password
|
v
ShopAuthContext.registerOwner()
|
+-- supabase.auth.signUp({ email, password, metadata })
|
+-- RPC register_shop_owner(...)
|   |
|   +-- INSERT shops (is_active=false, owner_user_id)
|   +-- INSERT shop_branches (slug=main)
|   +-- UPDATE users SET role=pending_owner, is_active=false, shop_id
|
v
Show: "Pending approval" (owner cannot access /shop yet)
|
v
[Admin reviews in /admin → Pending tab]
```

24.2 Admin approval flow

```
Admin logs in (admin@demo.com) via Owner tab on /welcome
|
v
AppBootstrap → router.replace('/admin')
|
v
AdminPanel → Pending requests tab
|
+-- [Accept] → RPC approve_shop_owner(user_id, shop_id)
|   |
|   +-- users: role=owner, is_active=true
|   +-- shops: is_active=true
|   |
|   v
|   Shop appears in customer catalog (refreshCatalog)
|   Owner can login → /shop
|
+-- [Reject] → RPC reject_shop_owner(user_id, shop_id)
|   |
|   +-- DELETE shop_branches + shops
|   +-- users: role=customer, shop_id=null
|   |
|   v
|   Registration removed; user stays as customer
```

24.3 Catalog visibility

```
refreshCatalog() queries shops WHERE is_active = true
|
```

```
v
pending_owner shops (is_active=false) NEVER shown to customers
|
v
After approve_shop_owner → shop included on next catalog refresh
```

24.4 Platform stats calculation

```
AdminPanel → Dashboard tab
|
v
fetchPlatformStats() in adminRepository.ts
|
+-- COUNT all bookings
+-- FILTER status = 'done' for revenue
+-- SUM platform_fee_egg OR 12% of service_price_egg
+-- COUNT pending_owner users
+-- COUNT active owner users
|
v
Display stat cards (OwnerSectionCard layout)
|
v
Merchant Ledger table (fetchMerchantLedger)
|
+-- Per shop: SUM platform_fee on done bookings AFTER platform_fee_last_settled_at
+-- [Settle Balance] → RPC admin_settle_shop_platform_fees → watermark = now()
```

24.5 Premium tier toggle (with confirmation)

```
Admin → Tab C Active Merchants → tap Premium switch
|
v
userConfirm() – "Update merchant subscription?" (EN/AR)
Web: window.confirm (OK = confirm, Cancel = abort)
Native: Alert.alert with two buttons
|
+-- Cancel → no change
+-- Confirm update (admin_premium_confirm_btn)
    |
    v
    RPC admin_toggle_shop_premium(shop_id, is_premium)
    |
    v
    shops.is_premium updated → owner /shop reflects instantly
    (PremiumFeatureGate unlocks or locks CMS sections)
|
v
userAlert() success toast
```

Do NOT use Alert.alert with buttons on web — buttons are invisible; toggle appears broken. Always use userConfirm for admin confirm flows.

24.6 Moderation queue

```
Sources of reports:
|
+-- Driver Network: user taps flag on PostCard → posts.reported = true
+-- Shop review: owner taps Report fake → shop_reviews.reported = true
|
v
```

```
Admin → Tab D Moderation → listModerationQueue()  
|  
+-- [Delete Content] → remove row from posts / comments / shop_reviews  
+-- [Dismiss Report] → reported = false (keep content)
```

24.7 Walk-in POS booking

```
Wash owner OR branch manager → /shop → Quick Counter Cashier (POS)  
|  
v  
WalkInBookingModal: car type (required) · phone optional + hint · service · notes  
NO plate · title "Quick Counter Cashier (POS)"  
|  
v  
createWalkInBooking() → INSERT bookings  
  booking_type = walk_in · status = confirmed (then owner completes → done)  
  service_price_egg from branch_services · platform_fee_egg = 12%  
  customer_id from resolve_customer_id_by_phone if phone matches app user  
|  
v  
Shows counter invoice (gross / fee / net) → live queue + ledger + owner PDF reports
```

25. Super Admin — Modules & UI

شرح وحدات لوحة المشرف

Route: `app/admin/index.tsx` + `components/admin/AdminPanel.tsx`

Full-screen layout (web-optimized): fixed header + horizontal tab chips + scrollable content.

Tab A — Dashboard (`admin_tab_dashboard`)

Stat card	Data source
Total bookings	COUNT bookings
Completed	status = done
Gross revenue	SUM service_price_egp (done)
Platform fee (12%)	SUM platform_fee_egp or calculated
Pending owners	COUNT users role=pending_owner
Active merchants	COUNT users role=owner, is_active

Below stats: Merchant Ledger table — outstanding 12% fees per shop + Settle Balance button.

Tab B — Pending requests (`admin_tab_pending`)

Each card shows: shop name, type, owner email, phone, address.

Accept → `approveShopOwner()` → RPC

Reject → `rejectShopOwner()` → RPC

Uses `userAlert` for success/error on web.

Tab C — Active merchants (`admin_tab_merchants`)

Lists approved owners: shop name, type, email, active branch count.

Premium Tier row: custom switch labeled `admin_premium_toggle_label`. Tap → `userConfirm(admin_premium_confirm_*)` → on OK → RPC `admin_toggle_shop_premium` → `userAlert success`.

Settle Balance (Tab A ledger) uses the same `userConfirm` pattern before `admin_settle_shop_platform_fees`.

Tab D — Moderation (`admin_tab_moderation`)

Unified queue: reported posts, comments, shop reviews. Actions: Delete Content / Dismiss Report.

Header actions

- Title: Super Admin / لوحة المشرف
- Signed-in email from `ShopAuthContext.staff`
- Sign out → `useAppSignOut()` → `/welcome`

26. Merchant Registration Module

وحدة تسجيل التاجر الجديد

UI: `app/welcome.tsx` — **Owner tab**

Toggle: Login ↔ Register as merchant (`owner_register_link`)

Register fields:

1. Business name (`shopName`)
2. Owner full name (`name`)
3. Egypt mobile phone — validated
4. Business address
5. Shop type chips: wash, maintenance, parts, accessories
6. Area chips: from catalog areas (Maadi, Nasr City, ...)
7. Email + password + **Re-enter password** (must match)

Submit: `registerOwner()` → success Alert → redirect to owner login tab. English-only fields; Arabic shop fields stay empty in DB unless provided (step 13).

Backend: `lib/shop/ownerRegistrationRepository.ts`

1. `supabase.auth.signUp` with metadata (`name`, `phone`, `account_type=shop_owner`)
2. `supabase.rpc('register_shop_owner', { p_shop_name, p_shop_type, p_area_id, ... })`
3. On RPC failure → `signOut` and return invalid
4. On success → **signOut** (no pending session flash); user logs in after admin approval

Login while pending

Credentials valid but `login()` returns `pending_approval`.

AppBootstrap keeps user on `/welcome` with pending message — no access to `/shop` or customer booking as owner.

27. New Library Modules (Admin)

ملفات الكود الجديدة

File	Exports / purpose
lib/admin/adminRepository.ts	listPendingOwnerRequests, listActiveMerchants, fetchPlatformStats, fetchMerchantLedger, listModerationQueue, toggleShopPremium, settleMerchantPlatformFees, dismissModerationReport, deleteModerationContent, approveShopOwner, rejectShopOwner
lib/shop/ownerRegistrationRepository.ts	registerShopOwner(input) → signUp + RPC
lib/shop/shopStaffUser.ts	fetchAppStaffUser, resolveShopSession — extended for admin + pending_owner
context/ShopAuthContext.tsx	isAdmin, isPendingOwner, registerOwner, pending_approval login result
components/AppBootstrap.tsx	Admin → /admin; pending → /welcome?pending=1
components/admin/AdminPanel.tsx	Full Super Admin UI
app/admin/_layout.tsx	Stack layout for admin route
supabase/apply-pitstop-2.0-step5-admin-approval.sql	Enum, RLS, RPCs, is_platform_admin()

Context flags quick reference

Flag	When true	Route
isAdmin	users.role = admin	/admin
isPendingOwner	users.role = pending_owner	/welcome (blocked)
isOwner	users.role = owner, active	/shop
isPremium	shops.is_premium = true for owner's shop	/shop — unlocks wash CMS sections
isBranchManager	users.role = branch_manager	/shop (branch filtered)

28. Car Wash Customer UX — Busy · Loyalty · Progress

مميزات تجربة عميل الغسيل — شارة المزدحم، الولاء، وتقديم الحجز

This section documents the three customer-facing Car Wash features added in the latest release, plus the auth fixes that unblock login on web. All three UX features work **offline-first** (AsyncStorage) and do not require new Supabase tables.

28.1 Dynamic Busy Status Indicator

Purpose: When a wash owner is overwhelmed, they set branch status to **Busy**. Customers see a clear red pulsing notice before booking — on the shop list and shop profile.

الغرض: يضبط صاحب المحل حالة «مزدحم» في لوحة التحكم، فيظهر للعميل تنبيه أحمر نابض في قائمة المحلات وصفحة المحل.

Owner side

1. Login as wash owner → /shop → WashOwnerPanel
2. Section **Branch status** (wash_status_* keys): choose Open / Closed / **Busy** / Vacation
3. Save → saveWashBranchStatus() writes to branch storage and calls syncWashBranchToShopExtras()

Data path

```
WashOwnerPanel.saveWashBranchStatus()
  → washBranchStorage (active branch)
  → syncWashBranchToShopExtras()
  → shopExtrasStorage (key per shopId)
  → field: washShopStatus = 'open' | 'closed' | 'busy' | 'vacation'
  → ShopCatalogContext loads extras for customer lists
```

Important fix: normalizeExtras() in shopExtrasStorage.ts now preserves washShopStatus, vacation dates, and activeBranchId — previously these fields could be stripped on save.

Customer UI

Screen	Component	Condition
Shop list card	WashBusyBadge compact	shop.type === 'wash' AND extras.washShopStatus === 'busy'
Shop profile	WashBusyBadge full width	Same condition

Visual: red border/background tint, animated opacity pulse on dot, exclamation icon, bilingual text.

i18n keys

Key	EN (summary)	AR (summary)
wash_busy_customer_notice	Shop is busy — expect longer wait	المحل مزدحم حالياً — قد يكون الانتظار أطول

How to test

1. Owner: set branch status to Busy → Save
2. Customer: open wash shop list in same area → red badge on that shop card
3. Open shop profile → full busy notice appears
4. Owner: set back to Open → badge disappears after catalog refresh

28.2 Offline-First Loyalty Stamps

Purpose: Reward repeat wash customers. Each completed wash booking adds one stamp; after **5 stamps** the customer receives a **100% off coupon** valid 90 days.

الغرض: برنامج ولاء — ختم واحد لكل غسيل مكتمل، وبعد 5 أختام كوبون خصم 100% لمدة 90 يوم.

Rules

- Only shopType === 'wash' bookings count
- Stamp added when booking status becomes done (not on create/cancel)
- Idempotent: each booking.id processed once (processedBookingIds array, max 200 kept)
- At 5 stamps: generate coupon, reset stamps to 0, show reward modal on Home
- Customer key: id:{customerId} or fallback phone:{phone}
- Guest users: no loyalty card (login required)

Storage

AsyncStorage key: @pitstop/loyalty_stamps

```
{
  "id:uuid-or-phone:012...": {
    "stamps": 3,
    "processedBookingIds": ["booking-1", "booking-2"],
    "pendingReward": null,
    "rewards": [{ "code": "LOYAL-ABC123", "discountType": "percent", "discountValue": 100, ... }]
  }
}
```

Code flow

Owner: updateBookingStatus(id, 'done')

- storage.ts calls recordWashBookingDone(booking, previousStatus)
- stamp +1 (or unlock reward at 5)

Customer: Home tab focus

- listBookingsForPhone()
- syncLoyaltyFromBookings() – backfill missed done washes
- Settings /points-marketplace for points balance

Reward coupon format

Field	Value
code	LOYAL- + 6 random uppercase chars
discountType	percent
discountValue	100 (free wash)
expiresAt	90 days from issue

Limitation: coupon is stored locally; booking checkout does not yet validate LOYAL-* codes automatically — future enhancement.

UI — Points marketplace (Settings)

- Route: /points-marketplace from Settings tab
- Shows loyalty points balance + partner shop cards
- Not shown on Home (legacy LoyaltyCard removed)

i18n keys

Key	Usage
loyalty_card_title	Card heading
loyalty_card_lead	Short program description

loyalty_card_progress	Stamp count ({count}, {goal})
loyalty_reward_title	Reward modal title
loyalty_reward_body	Modal body with {code}
loyalty_reward_dismiss	Close button

How to test

1. Login as customer@demo.com
2. Book a wash → owner marks booking Done (5 times, or change dates/status in storage for demo)
3. Home tab: stamp count increases after each done
4. On 5th stamp: modal shows LOYAL-XXXXXX, stamps reset to 0
5. Re-open Home: same booking does not add duplicate stamp

28.3 Live Queue Progress Bar (My Bookings)

Purpose: For *today's* active appointments, show a **3-step** horizontal timeline: Pending → Confirmed → Done.
 الغرض: للحجوزات اليوم فقط — 3 خطوات: قيد الانتظار → مؤكد → تم.

When shown

```
showTimeline =
  isBookingToday(scheduledAt)
  AND status IN (pending, confirmed, in_progress, done)
  // in_progress counts as step 2 (Confirmed) in UI
```

Cancelled / no-show bookings keep the normal colored status badge only.

Steps (left → right)

Step	Booking status	Visual
1	pending	First circle active
2	confirmed or legacy in_progress	Connector + circle 2 active
3	done	All steps accent-colored with check icons

Component: BookingProgressTimeline.tsx — uses theme accent for reached steps, muted for future steps.

i18n keys

Key	Step label
booking_progress_title	Section title above timeline
booking_progress_pending	Pending
booking_progress_confirmed	Confirmed
booking_progress_done	Done

booking_progress_in_progress remains in i18n for legacy data but is not a separate timeline step.

How to test

1. Customer books wash for **today**
2. My Bookings: timeline visible, step 1 active (pending)
3. Owner confirms → customer refreshes tab → step 2 active
4. Owner completes → step 3 (done); optional Rate footer appears

5. Book for tomorrow → only status badge, no timeline

28.4 Auth fixes (prerequisite for all roles)

These infrastructure + code fixes were required before customer/owner/admin login worked reliably on web:

Issue	Fix
Email provider disabled in Supabase	Dashboard → Auth → Providers → enable Email
Demo accounts unconfirmed	Auto-confirm on create OR SQL update email_confirmed_at
Web login stuck "Signing in..."	authMutationLock.ts + defer onAuthStateChange work
Browser tabs syncing auth	webTabAuthIsolation.ts — see §31
Admin not routing to panel	ok_admin result → /admin from welcome

Full checklist: supabase/SETUP-AUTH-CHECKLIST.md

إصلاحات المصادقة: تفعيل Email في Supabase، تأكيد الحسابات التجريبية، ومنع تعليق signIn على الويب.

28.5 File reference (Car Wash UX)

File	Role
components/ui/WashBusyBadge.tsx	Customer busy notice UI
components/ui/ShopListCard.tsx	Shows compact busy badge
app/shop-profile/[shopId].tsx	Shows full busy badge
lib/booking/shopExtrasStorage.ts	Persists washShopStatus
components/customer/ActiveVehiclePicker.tsx	Home vehicle picker
components/customer/OrderListCard.tsx	My Bookings + rating
app/points-marketplace.tsx	Loyalty points UI
components/customer/BookingProgressTimeline.tsx	3-step progress UI
lib/booking/loyaltyStampsStorage.ts	Stamp logic on booking done
lib/booking/storage.ts	Calls recordWashBookingDone on status update
lib/booking/reviewRepository.ts	One review per shop + orderHistoryReviewBody
app/(tabs)/bookings.tsx	OrderListCard + timeline + rating
lib/auth/authMutationLock.ts	Web auth deadlock guard
lib/i18n/strings.ts	All EN/AR strings above

29. SaaS Premium — Free vs Premium (Wash Owner)

اشتراك Premium للمغسلة — مجاني مقابل مدفوع

Wash owners use a **Freemium SaaS** model. Flag: `public.shops.is_premium` → `ShopAuthContext.isPremium`. SQL: `apply-pitstop-2.0-step6-shop-premium.sql`.

29.1 Enable Premium

Option A — Super Admin (recommended): Login `admin@demo.com` → `/admin` → Tab C → toggle Premium → confirm dialog → instant update.

Option B — SQL (manual):

```
update public.shops set is_premium = true where id = 'shop-wash-nile';
```

29.2 Free vs Premium

Free ✓	Premium only (locked with gold badge)
Today + pending stats, branch status, profile, gallery, GPS, bookings	Multi-branch tabs, analytics, services, hours, coupons, managers, employees, reviews, PDF reports, extra dashboard stats

29.3 Flow

```
Owner /shop → isPremium?  
No → PremiumFeatureGate dims sections → tap → PremiumUpgradeModal  
Yes → full WashOwnerPanel CMS + branch switcher
```

30. Branch GPS & Customer Location Flow

موقع الفرع بالـ Owner — GPS يلتقط Customer → يشوف المسافة والاتجاهات

Owner: Profile → Set Branch Location via GPS

- `saveBranchCoordinates()` → `shop_branches` + shops updated in Supabase
- Customer: `/nearby/wash` reloads (`useFocusEffect`) → Haversine sort
- Customer: shop-profile → Directions → Google Maps `?q=lat,lng`

30.1 El Rehab demo

1. Auth user `rehab.wash@demo.com / demo123`
2. Run `seed-el-rehab-wash.sql` — coords 30.0244, 31.4939
3. Owner captures GPS → customer sees updated distance in Nearby

Modules: `washBranchNearby.ts`, `branchRepository.ts`, `nearby.ts`, `contact.ts`, `washBranchStorage.saveBranchCoordinates`.

31. Web Multi-Tab Auth Isolation

كل تاب متصفح له session مستقل — owner في تاب و customer في تاب

- sessionStorage for Supabase JWT + app session keys
- Disable Supabase BroadcastChannel before createClient
- signOut({ scope: 'local' }) on web logout

Test: Tab1 owner → duplicate → Tab2 customer login → Tab1 stays owner. Hard refresh after code deploy.

32. Wash Pricing — Services Only (No Base Price)

أسعار الغسيل من قائمة الخدمات فقط — بدون سعر أساسي

Removed owner "Base price" from WashOwnerPanel. Customers see per-service prices on shop profile. Booking uses `selectedService.priceEgp` or first active wash service. Maintenance shops still use default service price in generic owner panel.

33. Walk-in POS — Quick Counter Cashier

تسجيل غسيل سريع (كاشير) — زبائن بدون تطبيق

```
Who: Wash owner OR branch manager (/shop shortcuts)
|
v
Modal: WalkInBookingModal.tsx
  Fields: car type/model (required) · phone (optional hint) · service menu · notes
  NO plate number (removed for speed)
  |
  v
Register & start wash → createWalkInBooking()
|
v
bookings row:
  booking_type = walk_in · status = confirmed (then owner completes → done)
  customer_phone nullable · customer_id if phone matches users.phone
  service_price_egp + platform_fee_egp (12%)
  |
  v
Counter invoice screen → refresh owner analytics + admin ledger
```

SQL: apply-pitstop-2.0-steps-7-through-8-bundle.sql + step7b RPC.

34. Driver Network & Community Feed

شبكة السائقين — منشورات، تعليقات، إعجابات

```
Customer tab: Driver Network (assistant.tsx)
|
+-- Create post (optimistic + Supabase posts table)
+-- Like posts (post_votes vote_type=up)
+-- Open post → nested comments + comment likes
+-- Flag post (reportCommunityPost → reported=true)
|
v
Tables: posts · comments · post_votes · comment_likes
Local AsyncStorage merge when offline
```

Deep links: pitstop://driver-network/[postId] · pitstop://shop-profile/[shopId]

35. Wash Owner vs Branch Manager (Role Split)

تقسيم الصلاحيات — مالك مقابل مدير فرع

Feature	Owner	Branch Manager
Branch tabs + Add branch	Premium only	Locked to assigned branch
Manager account provisioning	Yes	Hidden
Financial PDF reports (gross/fee/net)	Owner + Premium	Hidden
Shop status + Walk-in POS	Yes	Yes
Employees CRUD	Yes	Yes (own branch)
Reviews moderation	Yes	Yes
Operational analytics (peak hour, top service, trend)	Yes	Yes
Weekly revenue stat	Premium	Hidden

Component: WashOwnerPanel.tsx gates via isOwner, isBranchManager, isPremium. PremiumFeatureGate unchanged for locked CMS sections.

36. Super Admin SaaS — Premium · Ledger · Moderation

لوحة المشرف — بريميوم، مستحقات، رقابة

Tab A Dashboard

Platform stats + Merchant Ledger + Settle Balance per shop

Tab B Pending

Approve / reject new merchants

Tab C Merchants

Premium toggle with confirm dialog before RPC

Tab D Moderation

Reported posts, comments, shop reviews

Ledger math: outstanding = SUM(platform_fee_egg) on status=done bookings where scheduled_at > shops.platform_fee_last_settled_at.

SQL: apply-pitstop-2.0-step9-admin-saas.sql

36.1 Premium confirm i18n keys

Key	EN	AR
admin_premium_confirm_title	Update merchant subscription?	تحديث اشتراك التاجر؟
admin_premium_confirm_enable_body	Enable Premium for {shop}? ...	Premium ل {shop}؟ ... تفعيل
admin_premium_confirm_disable_body	Disable Premium for {shop}? ...	Premium ل {shop}؟ ... إلغاء
admin_premium_confirm_btn	Confirm update	تأكيد التحديث

36.2 Troubleshooting admin on web

Symptom	Cause	Fix
Premium toggle does nothing	Alert.alert buttons invisible on web	Use userConfirm (fixed in AdminPanel)
RPC error after confirm	Step 9 SQL not run	Run apply-pitstop-2.0-step9-admin-saas.sql
Owner still free after toggle ON	Owner session cached	Owner reload /shop; check shops.is_premium in Supabase

37. Branch Profile & Cover Photos

صورة البروفايل منفصلة عن صورة الغلاف

```
profileImageUrl → circular avatar only (OwnerProfileHeader)
imageUrls[0] → cover/banner only
imageUrls[1..] → gallery album
|
v
setShopProfileImage() – never writes profile into cover slot
setShopCoverImage() – replaces index 0 only
addShopImage() – appends gallery (skips profile URL duplicate)
```

38. SQL Migrations Steps 7–13 (Checklist)

ترتيب تشغيل SQL على Supabase — خطوة بخطوة

1. Ensure steps 1–6 already applied (see §9)
2. Run `apply-pitstop-2.0-steps-7-through-8-bundle.sql` — walk-in POS, reviews, community feed
3. Run `apply-pitstop-2.0-step9-admin-saas.sql` — ledger watermark, moderation, admin RPCs
4. Run `apply-pitstop-2.0-step10-customer-delete-account.sql` — `customer_delete_own_account()` RPC
5. Run `apply-pitstop-2.0-step11-offers.sql` — offers table + booking offer linkage
6. Run `apply-pitstop-2.0-step12-wash-loyalty.sql` — `customer_wash_loyalty` + Supabase stamp sync
7. Run `apply-pitstop-2.0-step13-owner-register-ar-empty.sql` — merchant register keeps `name_ar/address_ar` empty unless provided
8. Run `apply-pitstop-2.0-step13-campaigns.sql` — extends offers with `offer_type`, `discount_value`, `required_wash_count`, `expires_at`; Realtime on offers
9. Enable Realtime on bookings (if not already): `ALTER PUBLICATION supabase_realtime ADD TABLE public.bookings;`
10. Verify bundle checks show `ok = true`; reload app → test POS, bookings, admin, register

Full auth + demo setup: `supabase/SETUP-AUTH-CHECKLIST.md`

39. Web Dialogs — userAlert, userConfirm & CustomConfirmModal

حوارات الويب — تأكيد Premium والتسوية وتأكيد No-Show

React Native Alert.alert with multiple buttons is **unreliable on web** (Expo web). PitStop uses layered dialog helpers:

lib/ui/userAlert.ts → AppDialogProvider

Function	Web	Native	Used for
userAlert(title, message?)	Custom modal via AppDialogProvider	Alert.alert (patched)	Success/error one-liners after actions
userConfirm(title, message, opts?)	Custom modal via AppDialogProvider	Alert.alert + Cancel/OK	Premium toggle, Settle Balance, destructive confirms

lib/ui/CustomConfirmProvider.tsx + components/common/CustomConfirmModal.tsx

High-fidelity destructive confirms — especially merchant **Mark as No-Show / Decline**. Replaces ugly window.confirm on web.

API	Behavior
showCustomConfirm({ title, message, confirmLabel, cancelLabel, destructive, onConfirm })	Opens themed modal; async onConfirm with loading spinner
promptMerchantNoShowOverride()	Wrapper in lib/booking/merchantBookingOverride.ts

Mounted in: app/_layout.tsx inside AppDialogProvider

Call sites: components/owner/OwnerHistoryPanel.tsx · components/owner/wash/WashOwnerPanel.tsx · app/shop/wash-owner-hub.tsx

```
AdminPanel.onTogglePremium(row)
|
v
userConfirm(
  admin_premium_confirm_title,
  admin_premium_confirm_enable_body | disable_body // {shop} replaced
  { confirmLabel: admin_premium_confirm_btn, cancelLabel: alert_cancel }
)
|
+-- false → return (switch unchanged)
+-- true  → executeTogglePremium → RPC → userAlert success
```

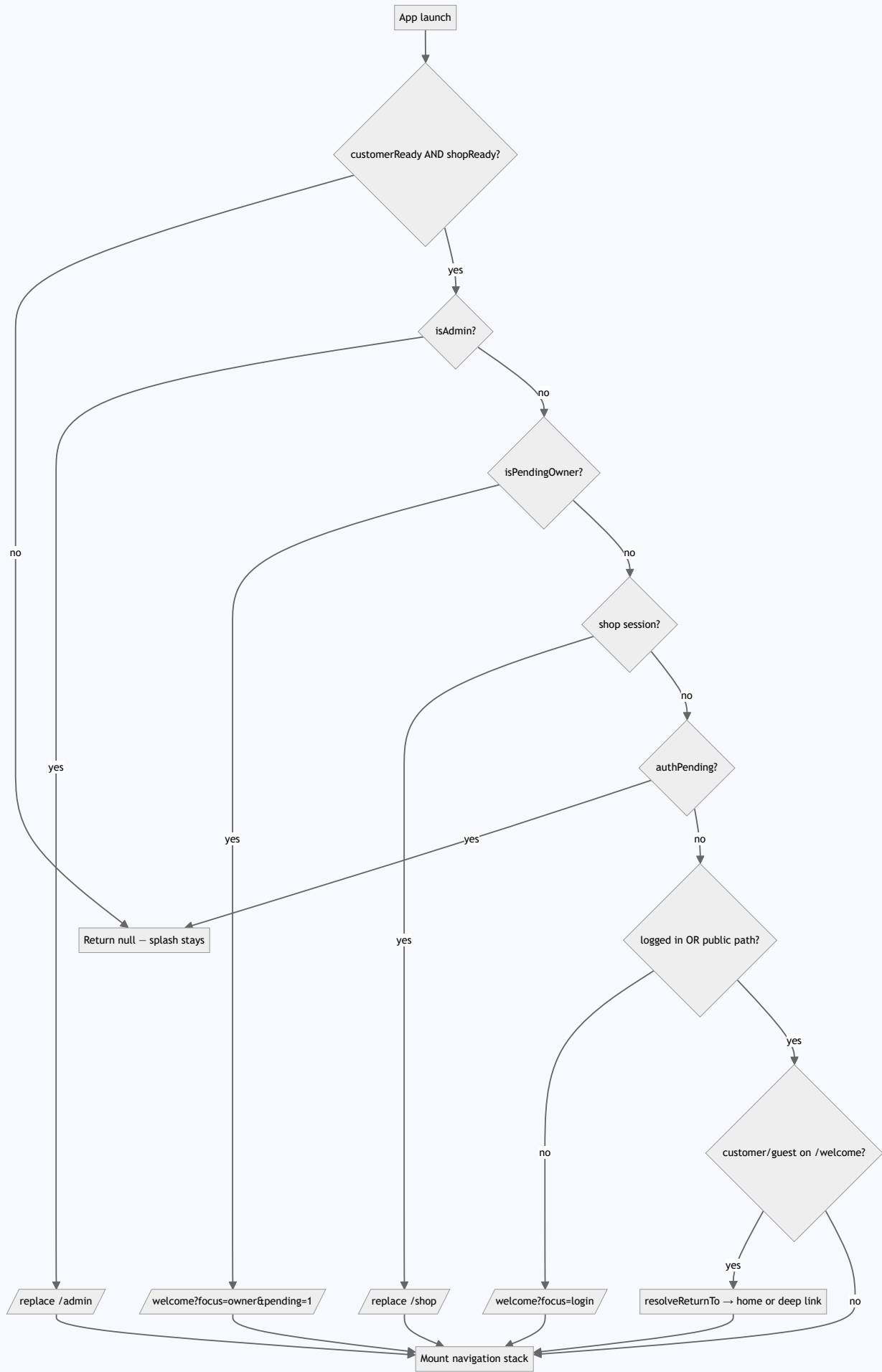
Files: components/admin/AdminPanel.tsx · lib/ui/userAlert.ts

40. Complete System Flowcharts (2026)

مخططات تدفق النظام الكامل — Jul 2026

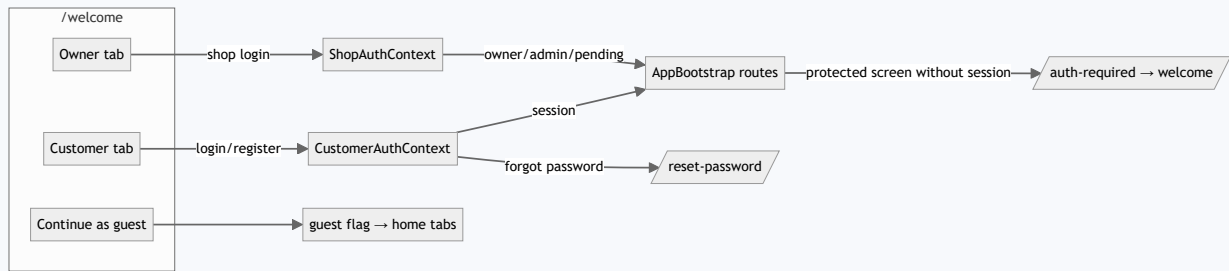
How to read: Interactive Mermaid diagrams render in a browser when you open this HTML file online or locally with internet (CDN). ASCII .flow boxes elsewhere in the doc still work offline / in PDF print.

40.1 App startup & routing (AppBootstrap.tsx)

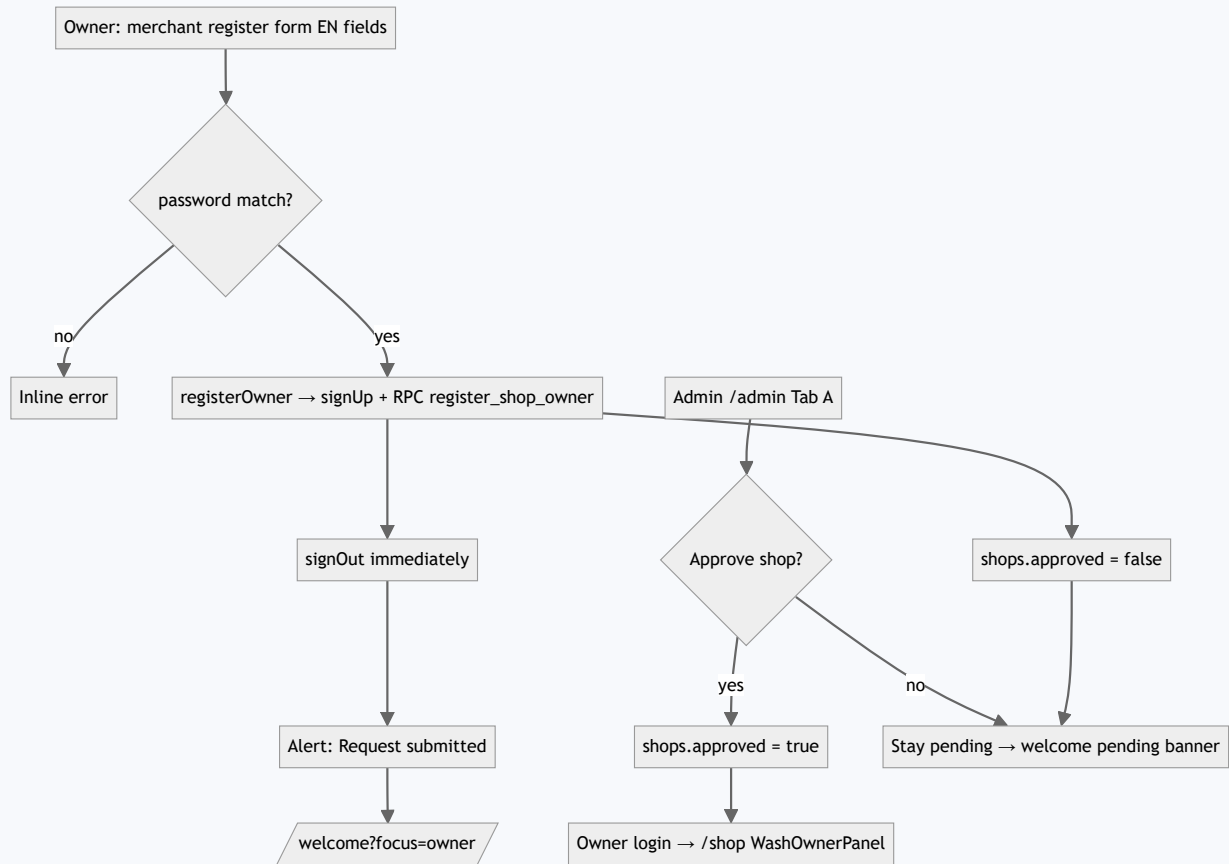


Public paths only: /welcome, /reset-password, /auth-required

40.2 Customer authentication

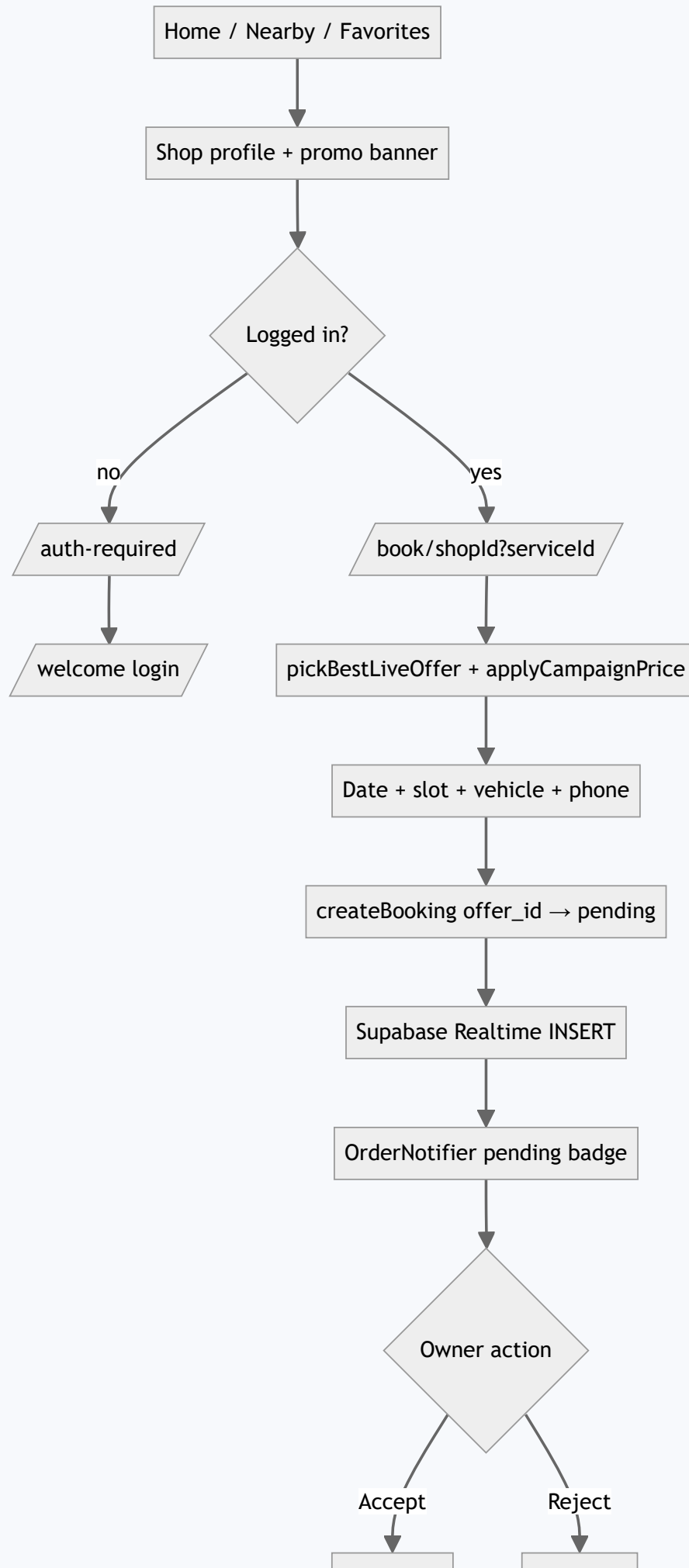


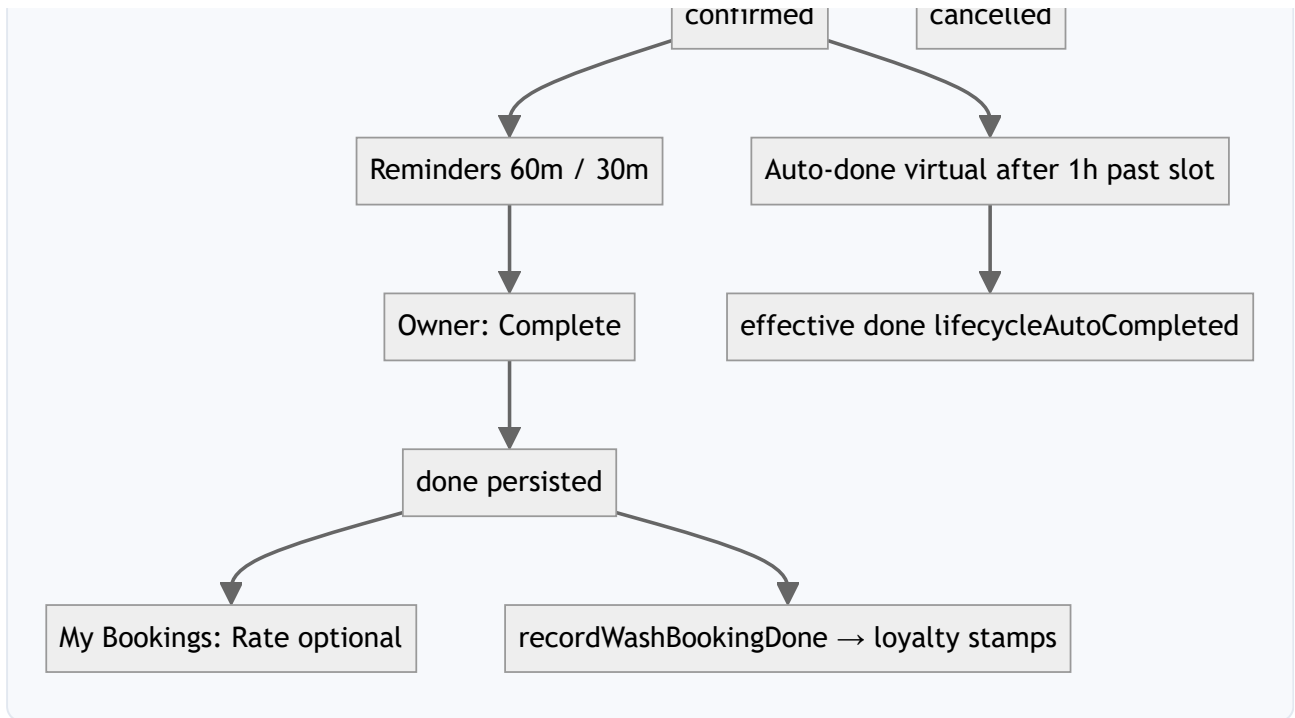
40.3 Merchant registration & approval



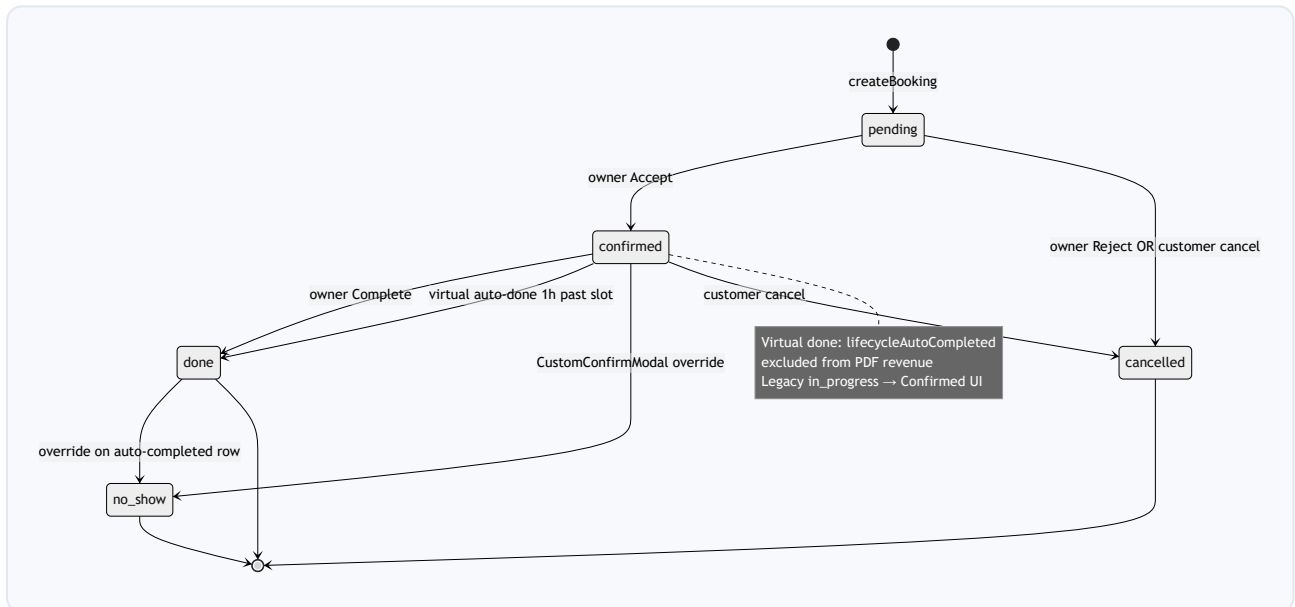
Step 13 SQL: Arabic name_ar / address_ar stored empty unless explicitly sent — no English copy fallback.

40.4 Customer booking journey

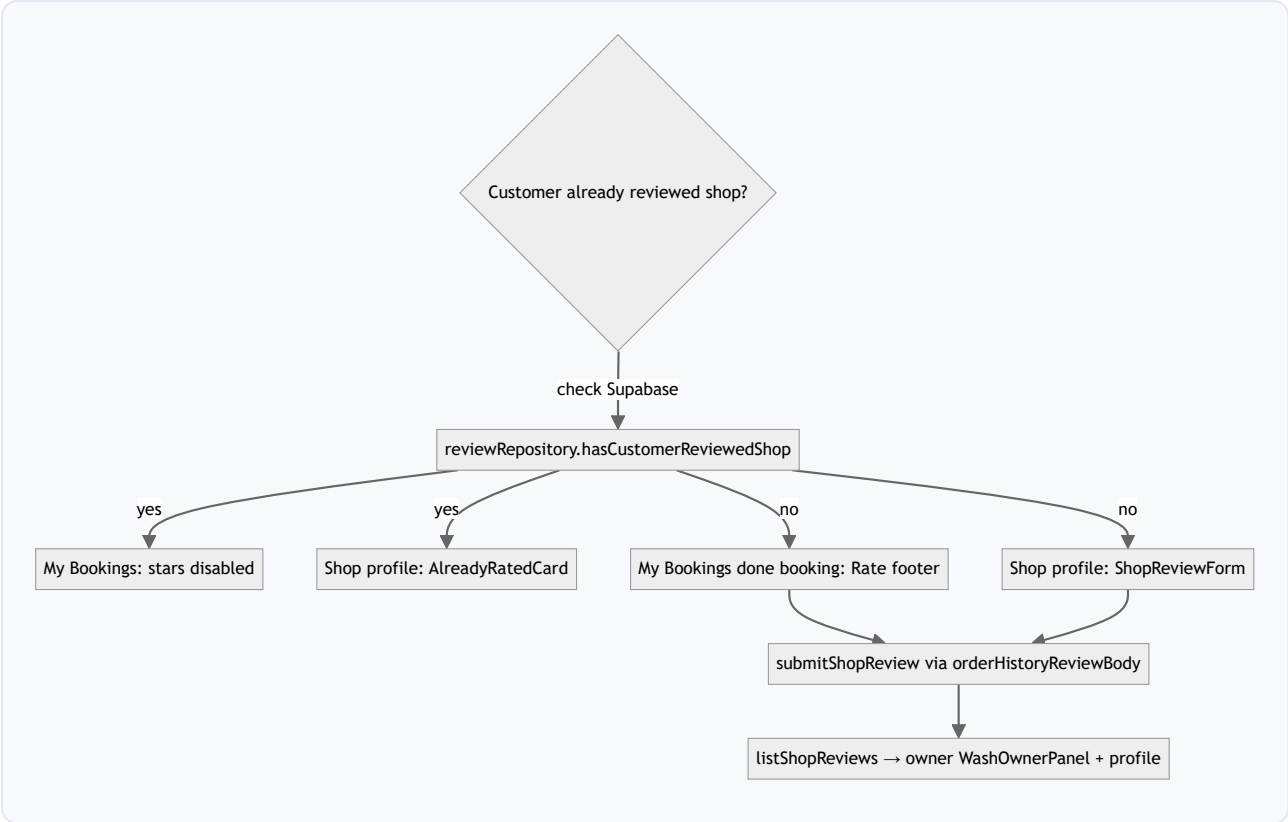




40.5 Booking status state machine (UI)

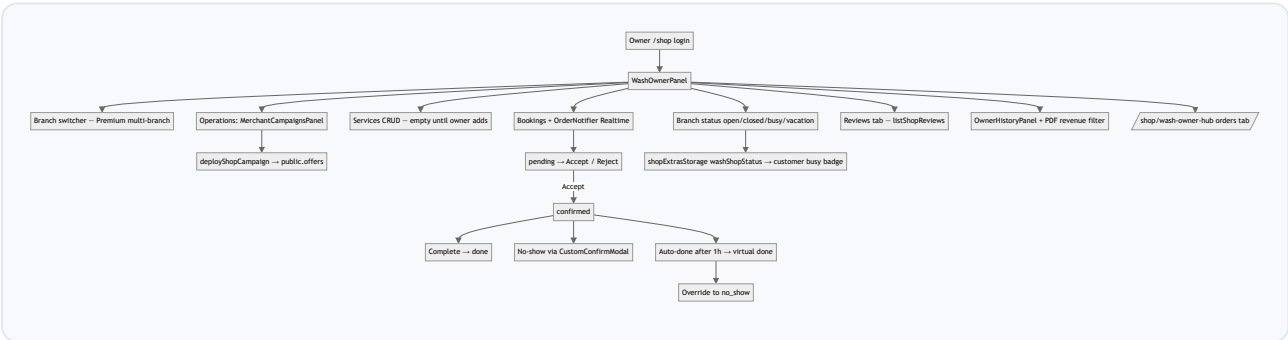


40.6 Reviews — one per customer per shop



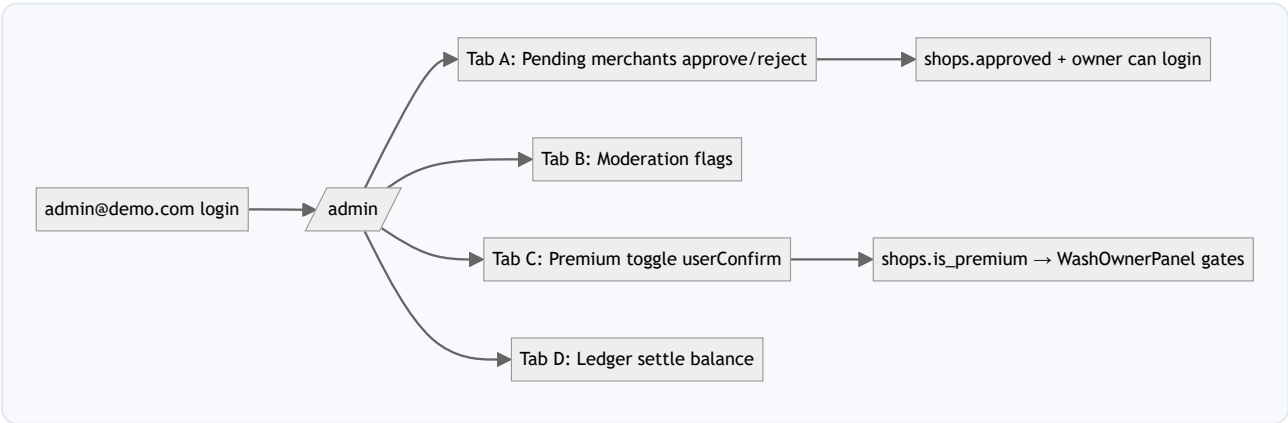
No dummy reviews: seedDemoReviews() removed — empty state when no real data.

40.7 Wash owner panel flow

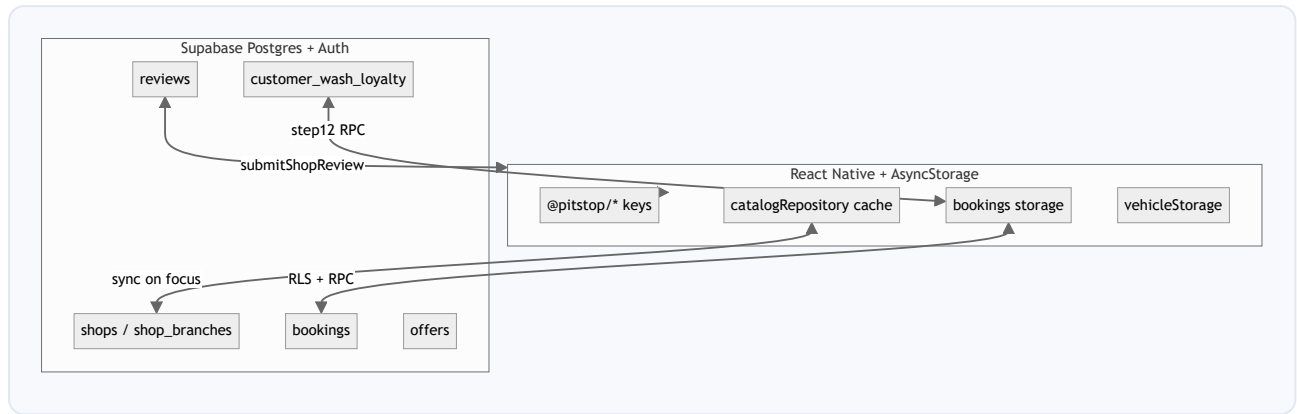


No "Publish default wash menu" — getActiveServices() returns [] until owner creates services.

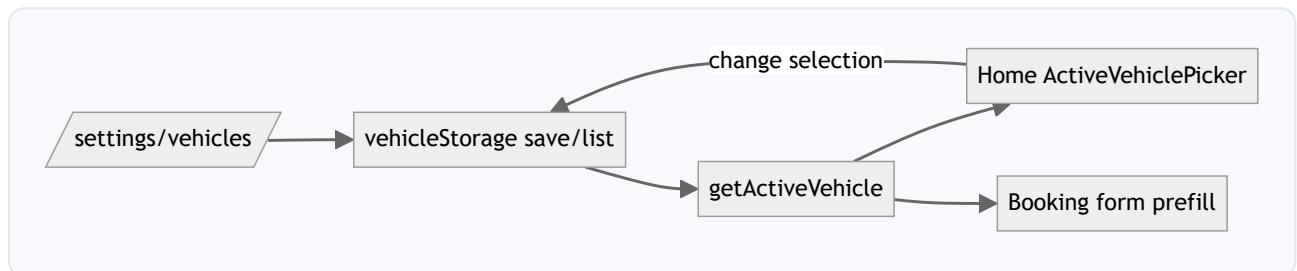
40.8 Platform admin



40.9 Hybrid data layer



40.10 Vehicles & active picker



40.11 Provider & screen map (summary)

```
ThemePreference → I18n → AppDialog → CustomConfirm → ShopCatalog → CustomerAuth → ShopAuth → AppBootstrap
|
+-- Customer tabs: Home | Bookings | Favorites | Assistant | Settings(My account)
+-- Stack: shop-profile, book, booking/[id], points-marketplace, admin, shop/*
+-- Owner: /shop (panel by type) + wash-owner-hub + merchant-settings stack + wash-history
```

41. Premium Splash & Home Hero Animation

شاشة البداية المتحركة ورسوم السيارة في الصفحة الرئيسية

41.1 Premium animated splash

File	Role
app/_layout.tsx	SplashScreen.preventAutoHideAsync(); overlay state showAnimatedSplash; hides native splash after fonts load
components/PremiumAnimatedSplash.tsx	Full-screen overlay bg #080D1A, accent #00D4FF, 1000ms hold + 500ms fade-out
components/AppBootstrap.tsx	Removed duplicate SplashScreen.hideAsync() — single hide path in layout

App launch → fonts load → SplashScreen.hideAsync (native)
→ PremiumAnimatedSplash visible 1.5s total
→ onFinish() → showAnimatedSplash = false
→ AppBootstrap routing proceeds

41.2 Home hero car animation

File	Role
components/home/HomeHeroCarAnimation.tsx	SVG car: idle translateY loop + periodic translateX drive; cleanup on blur via useFocusEffect
app/(tabs)/index.tsx	Hero row beside greeting for logged-in customers; offers carousel below service cards

42. Merchant Realtime & Order Notifier

إشعارات الطلبات الفورية للتاجر عبر Supabase Realtime

42.1 Architecture

File	Role
lib/notifications/notificationService.ts	Channel public:bookings:{shopId}; mapBookingRowFromRemote; mergeBookingList; applyVirtualBookingLifecycleBatch
components/merchant/OrderNotifier.tsx	Hook useMerchantOrderNotifier: pendingBookings, pendingCount, patchBookingLocally, refresh
components/owner/wash/WashOwnerPanel.tsx	orderNotifier integrated; notification badge on dashboard shortcuts
app/shop/wash-owner-hub.tsx	Orders tab uses orderNotifier.pendingBookings
app/(tabs)/shop.tsx	Generic shop owner pending count via orderNotifier

42.2 Realtime flow

```
Customer createBooking() → Supabase INSERT bookings (status=pending)
|
v
postgres_changes INSERT on public.bookings filter shop_id=eq.{shopId}
|
v
notificationService.handleRow → bookingEligibleForStaffAlert?
|-- branch_manager: only matching branch_id
|-- owner: all shop pending
v
onPendingInsert → triggerMerchantOrderAlert (Vibration + userAlert)
→ upsertPendingBookingSorted → pendingCount badge updates
|
v
Owner Accept/Complete → UPDATE bookings
→ onBookingUpdate → mergeBookingList + patchBookingLocally clears pending
```

42.3 Sorting fix (display order)

lib/booking/storage.ts → sortBookingsByScheduledAtDesc(): Tier 1 active (pending/confirmed/in_progress) above Tier 2 finalized; within tier newest scheduled_at first. Applied in listBookingsForShop, notification merges, customer My Bookings.

43. Smart Booking Lifecycle & No-Show Override

محرك دورة حياة الحجز التلقائي وتجاوز عدم الحضور

43.1 Virtual auto-done (read-time, not persisted)

Function	File	Behavior
AUTO_DONE_AFTER_MS	lib/booking/storage.ts	3600000 ms (1 hour grace after scheduled_at)
resolveEffectiveBookingStatus()	lib/booking/storage.ts	confirmed/in_progress + past window → effective status done
applyVirtualBookingLifecycle()	lib/booking/storage.ts	Sets lifecycleAutoCompleted: true on virtual transition
isAutoCompletedBooking()	lib/booking/storage.ts	Checks lifecycleAutoCompleted flag
isFinalizedHistoryBooking()	lib/booking/storage.ts	done cancelled no_show auto-completed
countsAsRevenueBooking()	lib/booking/storage.ts	Only persisted done without lifecycleAutoCompleted

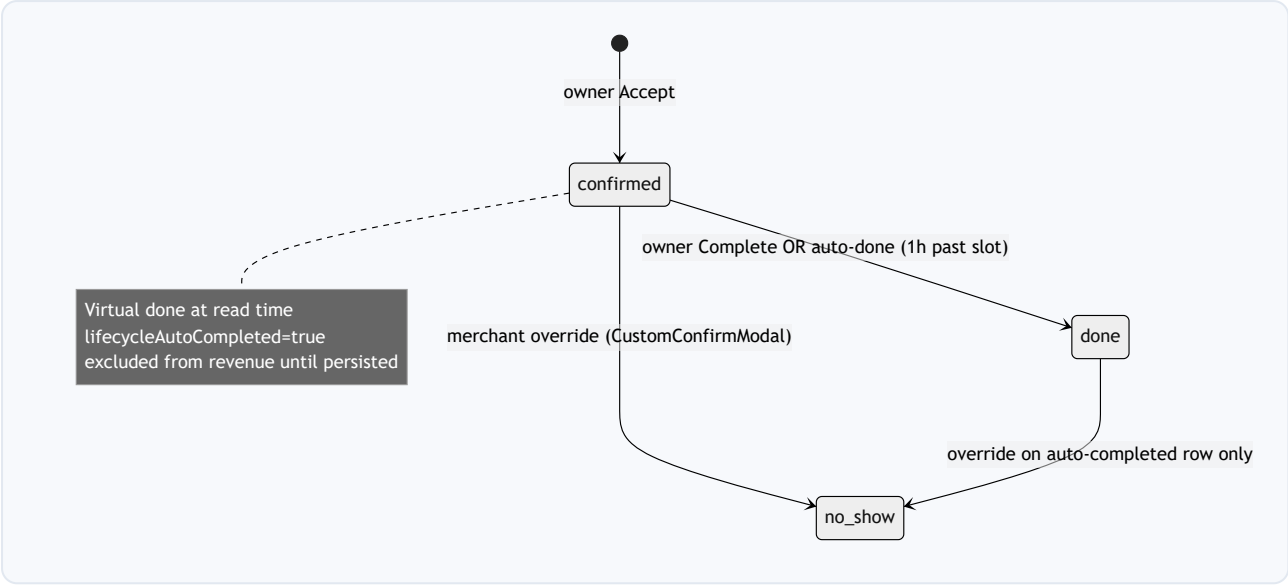
43.2 Merchant no-show override flow

```
OwnerHistoryPanel / WashOwnerPanel / wash-owner-hub
→ row has lifecycleAutoCompleted OR manual no-show action
→ promptMerchantNoShowOverride() [CustomConfirmModal]
→ onConfirm:
  updateBookingStatus(id, 'no_show') // persists to Supabase
  patchBookingLocally(id, 'no_show') // OrderNotifier instant UI
  setRows / refreshAll // history list updates
→ filterRevenueBookings excludes no_show from PDF totals immediately
```

43.3 Files

- lib/booking/merchantBookingOverride.ts — promptMerchantNoShowOverride
- components/owner/OwnerHistoryPanel.tsx — history filters, override button, reportBookings revenue filter
- lib/booking/bookingHistoryRepository.ts — listArchivedBookingsForStaff + sortArchivedBookingsForDisplay
- app/shop/wash-history.tsx — OwnerHistoryPanel mode=history
- lib/booking/reporting.ts — filterRevenueBookings(); PDF only counts status=done without auto-completed flag
- components/customer/OrderListCard.tsx — status badges; book again hidden for pending/confirmed
- lib/booking/customerOrderPresentation.ts — in_progress normalized to confirmed for customer display

43.4 Lifecycle state diagram



44. Custom Confirm Modal (Web + Native)

نافذة تأكيد مخصصة بدلاً من window.confirm

File	Role
components/common/CustomConfirmModal.tsx	Presentational Modal: rgba(0,0,0,0.7) backdrop, navy card, muted Cancel + solid red Confirm
lib/ui/CustomConfirmProvider.tsx	Global queue; showCustomConfirm(); mounted in app/_layout.tsx
lib/booking/merchantBookingOverride.ts	promptMerchantNoShowOverride → showCustomConfirm(destructive: true)

User taps "Mark as No-Show / Decline"

- showCustomConfirm({ title: merchant_noshow_override_title, ... })
- CustomConfirmModal visible (fade animation)
- Cancel → close modal, no DB write
- Confirm → await onConfirm() → updateBookingStatus('no_show') → close modal

i18n keys: merchant_noshow_override_title, merchant_noshow_override_body, merchant_noshow_override_btn
(EN + AR in lib/i18n/strings.ts)

45. Marketing Offers & Campaigns Engine

محرك العروض والحملات التسويقية

45.1 Database — public.offers (extended Step 13)

Migration: supabase/apply-pitstop-2.0-step13-campaigns.sql

Column	Type	Purpose
id	uuid PK	Offer identity; linked from bookings.offer_id
shop_id	text FK → shops.id	Merchant scope
title, description	text	Campaign copy
offer_type	text	percentage flat_amount buy_x_get_y
discount_value	numeric	20 = 20% or 50 EGP
required_wash_count	int	Buy-X threshold (e.g. 2 for buy 2 get 1 free)
is_active	boolean	Merchant toggle via deactivateShopOffer
expires_at	timestamp tz	Campaign end; nullable = open-ended
discount_percentage, start_date, end_date	legacy	Backward compatibility; synced on deploy

RLS: Anyone read live offers; shop staff CRUD via can_manage_shop(shop_id). **Realtime:** publication supabase_realtime includes offers.

45.2 Merchant deploy flow

```
WashOwnerPanel Operations tab OR shop.tsx workspace
→ MerchantCampaignsPanel
  MerchantCampaignForm: title, description, offer_type chips, value fields, valid days
  → deployShopCampaign() in lib/booking/offerRepository.ts
  → Supabase INSERT public.offers
  → Realtime broadcast
  MerchantCampaignList: active rows + "End campaign" → deactivateShopOffer()
```

Component	Path
MerchantCampaignForm	components/merchant/MerchantCampaignForm.tsx
MerchantCampaignList	components/merchant/MerchantCampaignList.tsx
MerchantCampaignsPanel	components/merchant/MerchantCampaignsPanel.tsx
Wash mount	components/owner/wash/WashOwnerPanel.tsx (adminTab=operations)
Shop mount	app/(tabs)/shop.tsx (workspace tab)

45.3 Customer discovery surfaces

Surface	File	Behavior
Home offers carousel	app/(tabs)/index.tsx	listAllActiveOffers + formatOfferBadge; subscribeOffersRealtime refresh
Shop list cards	components/ui/ShopListCard.tsx	pickBestLiveOffer + warmSoft promo capsule
Shop profile header	app/shop-profile/[shopId].tsx	promoBanner with badge + title + description

Shop extras cache	lib/booking/shopExtrasStorage.ts	getShopExtras merges listActiveOffersForShop from Supabase
-------------------	----------------------------------	--

45.4 Checkout pricing (app/book/[shopId].tsx)

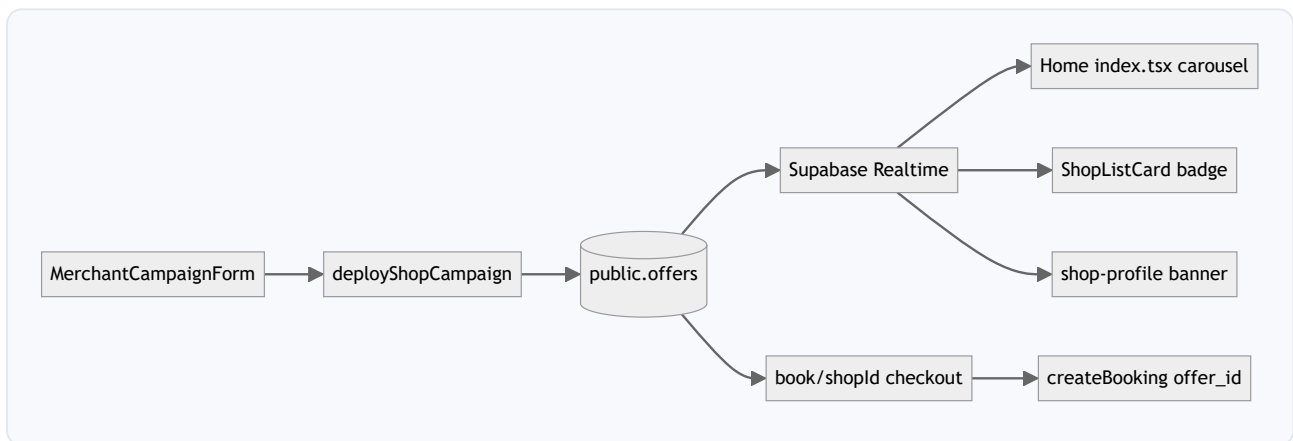
```

pickBestLiveOffer(shopExtras.offers)
|
+-- percentage → applyCampaignPrice: base * (1 - value/100)
+-- flat_amount → base - value (min 0)
+-- buy_x_get_y → countDoneBookingsForCustomerAtShop(status=done)
    cycle = required_wash_count + 1
    if (doneCount + 1) % cycle === 0 → price = 0 EGP (FREE wash banner)
|
v
createBooking({ offerId, servicePriceEgp: final }) via lib/booking/storage.ts
→ validateOfferForBooking + applyCampaignPrice on server path too

```

Pricing lib: lib/booking/offerPricing.ts — applyCampaignPrice, formatOfferBadge, buildOfferBadgeMessages, isBuyXGetYFreeNext

45.5 Campaign deploy diagram



46. Merchant History, Reports & Financial Integrity

سجل التاجر والتقارير المالية وسلامة الإيرادات

46.1 History panel

File	Role
components/owner/OwnerHistoryPanel.tsx	Filters: All / Completed / Cancelled; PDF date range; financial totals
lib/booking/bookingHistoryRepository.ts	listArchivedBookingsForStaff; only finalized + auto-completed
app/shop/wash-history.tsx	Wash history screen wrapper
app/(tabs)/shop.tsx	History tab via OwnerHistoryPanel variant=shop
app/shop/wash-reports.tsx	Reports-only mode

46.2 Financial integrity rules

- `filterRevenueBookings()` in `lib/booking/reporting.ts` — only status === 'done' without `lifecycleAutoCompleted`
- Auto-completed virtual rows excluded from PDF gross/fee/net until merchant confirms or overrides
- `no_show` and `cancelled` never count toward revenue
- On no-show override: instant local state via `patchBookingLocally` + `setRows` — no manual refresh required

46.3 Customer My Bookings UX

File	Change
app/(tabs)/bookings.tsx	Uses <code>OrderListCard</code> with status-colored badges
components/customer/OrderListCard.tsx	Rate block only for done; book again hidden for pending/confirmed
lib/booking/storage.ts	<code>sortBookingsByScheduledAtDesc</code> on customer fetch paths